Entanglement for a Quenched Harmonic Chain

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Sections 1-6 are calculations reproduced from [1]

1 General Formalism for Time Dependent Harmonic Oscillator

Let $H(t)$ be any general hamiltonian which depends on time explicitly. It is no longer a conserved quantity in the system. Hence, there are no quantum numbers corresponding to the hamiltonian anymore. Now, assume that there is a conserved quantity in the system, denote by $I(t)$. Then,

$$\frac{\mathrm{d}I}{\mathrm{d}t} = \frac{\partial I}{\partial t} + \frac{1}{i\hbar}[I, H] = 0 \tag{1.1}$$

Assuming that $I(t)$ is an observable. Hence $I^\dagger = I$. I then will have real eigen values. The Schrodinger equation is,

$$i\hbar\frac{\partial}{\partial t}\left|\Psi(t)\right\rangle = H(t)\left|\Psi t\right\rangle \tag{1.2}$$

Use the same ket to equation 1.1,

$$\begin{aligned}
i\hbar\frac{\partial I}{\partial t}\left|\Psi(t)\right\rangle + [I, H]\left|\Psi(t)\right\rangle &= 0 \\
i\hbar\frac{\partial I}{\partial t}\left|\Psi(t)\right\rangle + IH\left|\Psi(t)\right\rangle - HI\left|\Psi(t)\right\rangle &= 0 \\
i\hbar\left[\frac{\partial}{\partial t}\left(I\left|\Psi(t)\right\rangle\right) - I\frac{\partial}{\partial t}\left|\Psi(t)\right\rangle\right] + IH\left|\Psi(t)\right\rangle - HI\left|\Psi(t)\right\rangle &= 0 \\
i\hbar\frac{\partial}{\partial t}\left(I\left|\Psi(t)\right\rangle\right) - HI\left|\Psi(t)\right\rangle + I\left[i\hbar\frac{\partial}{\partial t}\left|\Psi(t)\right\rangle - H(t)\left|\Psi t\right\rangle\right] &= 0 \\
i\hbar\frac{\partial}{\partial t}\left(I\left|\Psi(t)\right\rangle\right) &= HI\left|\Psi(t)\right\rangle
\end{aligned}$$

Which means that $I\left|\Psi(t)\right\rangle$ is another solution of the Schrodinger equation. If I do not involve a time derivative, it is possible to construct eigen states of I such that they solve the Schrodinger equation. For that, assume that λ is the eigen value of I, and k represents the collective index which is required to specify a state completely and uniquely. Then,

$$I(t)\left|\lambda, k\right\rangle = \lambda\left|\lambda, k\right\rangle \tag{1.3}$$

Such that, $\{\left|\lambda, k\right\rangle : \forall\lambda \text{ and } k\}$ forms a complete orthonormal basis for the system. Also,

$$\left\langle\lambda', k'|\lambda, k\right\rangle = \delta_{\lambda\lambda'}\,\delta_{kk'} \tag{1.4}$$

The eigen values λ are time independent. For proving it, start from 1.1 and act it upon some eigen state of $I(t)$,

$$\begin{aligned}
i\hbar\frac{\partial I}{\partial t}\left|\lambda, k\right\rangle + IH\left|\lambda, k\right\rangle - HI\left|\lambda, k\right\rangle &= 0 \\
\Rightarrow i\hbar\left\langle\lambda', k'\right|\frac{\partial I}{\partial t}\left|\lambda, k\right\rangle + (\lambda' - \lambda)\left\langle\lambda', k'\right|H\left|\lambda, k\right\rangle &= 0 \\
\Rightarrow \left\langle\lambda, k'\right|\frac{\partial I}{\partial t}\left|\lambda, k\right\rangle &= 0
\end{aligned}$$

Now, start with 1.3 and differentiate it w.r.t. time,

$$\frac{\partial I}{\partial t}|\lambda,k\rangle + I\frac{\partial}{\partial t}|\lambda,k\rangle = \frac{\partial \lambda}{\partial t}|\lambda,k\rangle + \lambda\frac{\partial}{\partial t}|\lambda,k\rangle$$
$$\langle\lambda',k'|\frac{\partial I}{\partial t}|\lambda,k\rangle = \langle\lambda',k'|\frac{\partial \lambda}{\partial t}|\lambda,k\rangle + (\lambda-\lambda')\langle\lambda',k'|\frac{\partial}{\partial t}|\lambda,k\rangle$$

When $\lambda = \lambda'$,

$$\frac{\partial \lambda}{\partial t} = \langle\lambda,k'|\frac{\partial I}{\partial t}|\lambda,k\rangle = 0 \tag{1.5}$$

As the eigen values are time independent, the eigen vectors must be time dependent. Then by differentiating 1.3 by time, we get,

$$(\lambda - I)\frac{\partial}{\partial t}|\lambda,k\rangle = \frac{\partial I}{\partial t}|\lambda,k\rangle$$

Multiply by $\langle\lambda',k'|$ from the left,

$$\begin{aligned}(\lambda-\lambda')\langle\lambda',k'|\frac{\partial}{\partial t}|\lambda,k\rangle &= \langle\lambda',k'|\frac{\partial I}{\partial t}|\lambda,k\rangle \\ &= (\lambda-\lambda')\langle\lambda',k'|\frac{H}{i\hbar}|\lambda,k\rangle\end{aligned}$$

For $\lambda \neq \lambda'$, $|\lambda,k\rangle$ solves the Schrodinger equation. We cannot say anything about the other case. If the same equation did hold for For $\lambda = \lambda'$ also, then we could say that the eigen states of $I(t)$ satisfy the Schrodinger eqn. But There is a way to get the eigenstates to satisfy the S.E. Define,

$$|\lambda,k\rangle_\alpha = e^{i\alpha_{\lambda k}(t)}|\lambda,k\rangle \tag{1.6}$$

Where $\alpha_{\lambda k}(t)$ is real. $|\lambda,k\rangle_\alpha$ is still an eigenstate of I as we have only added a phase factor. We have for $\lambda \neq \lambda'$,

$$i\hbar\,\langle\lambda',k'|_\alpha\frac{\partial}{\partial t}|\lambda,k\rangle_\alpha = \langle\lambda',k'|_\alpha\frac{H}{i\hbar}|\lambda,k\rangle_\alpha \tag{1.7}$$

Which gives,

$$i\hbar\langle\lambda',k'|e^{-i\alpha_{\lambda k}}i\frac{\mathrm{d}\alpha_{\lambda k}}{\mathrm{d}t}e^{i\alpha_{\lambda k}}|\lambda,k\rangle + i\hbar\langle\lambda',k'|e^{-i\alpha_{\lambda k}}e^{i\alpha_{\lambda k}}\frac{\partial}{\partial t}|\lambda,k\rangle = \langle\lambda',k'|e^{-i\alpha_{\lambda k}}He^{i\alpha_{\lambda k}}|\lambda,k\rangle$$
$$\Rightarrow \qquad \hbar\frac{\mathrm{d}\alpha_{\lambda k}}{\mathrm{d}t}\delta_{\lambda\lambda'}\delta_{kk'} = \langle\lambda',k'|\left(i\hbar\frac{\partial}{\partial t} - H\right)|\lambda,k\rangle \tag{1.8}$$

Demand the above to be true even for $\lambda = \lambda'$. Then,

$$\hbar\frac{\mathrm{d}\alpha_{\lambda k}}{\mathrm{d}t}\delta_{kk'} = \langle\lambda,k'|\left(i\hbar\frac{\partial}{\partial t} - H\right)|\lambda,k\rangle$$

The operator $i\hbar\frac{\partial}{\partial t} - H$ should be diagonelized in such a way that it vanish for $k \neq k'$. This can always be done as $i\hbar\frac{\partial}{\partial t} - H$ is hermitian. Then, the phase should obey the equation,

$$\hbar\frac{\mathrm{d}\alpha_{\lambda k}}{\mathrm{d}t} = \langle\lambda,k|\left(i\hbar\frac{\partial}{\partial t} - H\right)|\lambda,k\rangle \tag{1.9}$$

Then the states defined by $|\lambda,k\rangle_\alpha$ are the eigenstates of $I(t)$ which solves the Schrodinger equation! Hence any general solution to the Schrodinger equation can be written as,

$$|\Psi(t)\rangle = \sum_{\lambda,k}c_{\lambda k}e^{i\alpha_{\lambda k}(t)}|\lambda,k;t\rangle \tag{1.10}$$

Where $c_{\lambda k}$ are time independent constants.

2 Transition Amplitudes

We now try to find out the transition amplitudes for a time dependent system. For that, we assume that the Hamiltonian settles into a constant value in the remote past ($t \to -\infty$) as well as in the distant future ($t \to \infty$). The hamiltonian will have a complete set of eigenstates in the remote past and the distant future, and denote it by $|n; i\rangle$ and $|m; f\rangle$ respectively. In the intermediate times, the hamiltonian has arbitrary time dependence although piecewise continuous. Assume that the system in the remote past started of with $|\Psi(-\infty)\rangle = |n; i\rangle$ and ends up in $|\Psi(\infty)\rangle = |m; f\rangle$. What is the transition amplitude, $T(n \to m)$?
We have the eigenstates of the invariant I denoted by $|\lambda, k; t\rangle$. Using them,

$$|n; i\rangle = \sum_{\lambda,k} c_{\lambda k} e^{i\alpha_{\lambda k}(-\infty)} |\lambda, k; -\infty\rangle \tag{2.1}$$

Which directly gives,

$$c_{\lambda k} = e^{-i\alpha_{\lambda k}(-\infty)} \langle\lambda, k; -\infty|n; i\rangle \tag{2.2}$$

Hence,

$$|\Psi(t)\rangle = \sum_{\lambda,k} e^{i(\alpha_{\lambda k}(t) - \alpha_{\lambda k}(-\infty))} |\lambda, k; t\rangle \langle\lambda, k; -\infty|n; i\rangle \tag{2.3}$$

We have chosen the state as $t \to \infty$ is $|m; f\rangle$. Hence,

$$T(n \to m) = \langle m; f|\Psi(\infty)\rangle = \sum_{\lambda,k} e^{i(\alpha_{\lambda k}(\infty) - \alpha_{\lambda k}(-\infty))} \langle m; f|\lambda, k; \infty\rangle \langle\lambda, k; -\infty|n; i\rangle \tag{2.4}$$

3 The Time Dependent Harmonic Oscillator

We start with a Hamiltonian which has a time dependent frequency,

$$H(t) = \frac{1}{2M}(p^2 + \Omega^2(t)q^2) \tag{3.1}$$

Where $\Omega(t)$ is the time dependent frequency. We assume that it is piecewise continuous and $\Omega^2(t)$ is real. Also,

$$[q, p] = i\hbar \tag{3.2}$$

We can write equations of motion as,

$$\dot{q} = \frac{1}{i\hbar}[q, H] = \frac{p}{M} \tag{3.3}$$

$$\dot{p} = \frac{1}{i\hbar}[p, H] = -\frac{\Omega^2(t)}{M} q \tag{3.4}$$

As the hamiltonian is a function of time, it is no longer a constant of motion. Now, we assume the existence of an invariant, which is quadratic in the dynamical variables. i.e,

$$I(t) = \frac{1}{2}[\alpha(t)q^2 + \beta(t)p^2 + \gamma(t)\{q, p\}] \tag{3.5}$$

Assume $I(t)$ to be an observable. Then $\alpha(t), \beta(t)$ and $\gamma(t)$ has to be real valued functions. As $I(t)$ is a constant of motion,

$$\frac{\mathrm{d}I(t)}{\mathrm{d}t} = \frac{\partial I(t)}{\partial t} + \frac{1}{i\hbar}[I, H] = 0 \tag{3.6}$$

Now,

$$\frac{\partial I}{\partial t} = \frac{1}{2}[\dot{\alpha}(t)q^2 + \dot{\beta}(t)p^2 + \dot{\gamma}(t)\{q,p\}]$$

And,

$$\frac{1}{i\hbar}[I,H] = \frac{1}{2}\left[\frac{\alpha}{M}\{q,p\} - \frac{\Omega^2\beta}{M}\{q,p\} + \frac{2\gamma}{M}p^2 - \frac{2\gamma\Omega^2}{M}q^2\right]$$

Hence,

$$\frac{\mathrm{d}I}{\mathrm{d}t} = \frac{1}{2}\left[\left(\dot{\alpha} - \frac{2\gamma\Omega^2}{M}\right)q^2 + \left(\dot{\beta} + \frac{2\gamma}{M}\right)p^2 + \left(\dot{\gamma} + \frac{\alpha}{M} - \frac{\beta\Omega^2}{M}\right)\{q,p\}\right] = 0 \tag{3.7}$$

For $\frac{\mathrm{d}I}{\mathrm{d}t}$ to be zero, each of the coefficients must go to zero. Then,

$$\begin{aligned} \dot{\alpha} &= \frac{2\gamma}{M}\Omega^2 \\ \dot{\beta} &= -\frac{2\gamma}{M} \\ \dot{\gamma} &= -\frac{\alpha}{M} + \frac{\beta}{M}\Omega^2 \end{aligned} \tag{3.8}$$

Do a substitution,

$$\beta(t) = \sigma^2(t) \tag{3.9}$$

Then from the second of equations 3.8,

$$\gamma = -M\sigma\dot{\sigma} \tag{3.10}$$

From the third of equations 3.8,

$$\alpha = M^2(\dot{\sigma}^2 + \sigma\ddot{\sigma}) + \Omega^2\sigma^2 \tag{3.11}$$

Now substitute this in the first of 3.8,

$$\begin{aligned} \dot{\alpha} = M^2(\sigma\dddot{\sigma} + 3\dot{\sigma}\ddot{\sigma}) + 2\Omega\dot{\Omega}\sigma^2 + 2\sigma\dot{\sigma}\Omega^2 &= \frac{2\gamma}{M}\Omega^2 \\ \Rightarrow \qquad M^2\sigma\dddot{\sigma} + 3M^2\dot{\sigma}\ddot{\sigma} + 2\Omega\dot{\Omega}\sigma^2 + 2\sigma\dot{\sigma}\Omega^2 + 2\sigma\dot{\sigma}\Omega^2 &= 0 \\ \Rightarrow \qquad \sigma(M^2\dddot{\sigma} + 2\Omega\dot{\Omega}\sigma + \Omega^2\dot{\sigma}) + \dot{\sigma}(3M^2\ddot{\sigma} + 3\Omega^2\sigma) &= 0 \\ \Rightarrow \qquad \sigma\frac{\mathrm{d}}{\mathrm{d}t}(M^2\ddot{\sigma} + \Omega^2\sigma) + 3\dot{\sigma}(M^2\ddot{\sigma} + \Omega^2\sigma) &= 0 \\ \Rightarrow \qquad \int\frac{d(M^2\ddot{\sigma} + \Omega^2\sigma)}{M^2\ddot{\sigma} + \Omega^2\sigma} &= -3\int\frac{d\sigma}{\sigma} \end{aligned}$$

The integral will give the constraint on σ as,

$$M^2\ddot{\sigma} + \Omega^2(t)\sigma - \frac{c}{\sigma^3} = 0 \tag{3.12}$$

Where c is the constant of integration. Using the above equation, we have,

$$\alpha = M^2\dot{\sigma}^2 + \frac{c}{\sigma} \tag{3.13}$$

Then the invariant can be calculated as,

$$
\begin{aligned}
I(t) &= \frac{1}{2}\left[\left(M^2\dot{\sigma}^2 + \frac{c}{\sigma}\right)q^2 + \sigma^2 p^2 - m\sigma\dot{\sigma}(qp+pq)\right] \\
\Rightarrow \qquad I(t) &= \frac{1}{2}\left[\frac{c}{\sigma^2}q^2 + (\sigma p - M\dot{\sigma}q)^2\right] \qquad (3.14)
\end{aligned}
$$

The constant of integration is a scalable quantity. We can, without loss of any generality, put it to unity. For that, take,

$$\sigma = c^{1/4}\rho$$

Then $I(t)$ become,

$$I(t) = \frac{\sqrt{c}}{2}\left[\frac{1}{\rho^2}q^2 + (\rho p - M\dot{\rho}q)^2\right]$$

AS $\sqrt{c}$ is just a constant, the invariant is,

$$I(t) = \frac{1}{2}\left[\frac{q^2}{\rho^2} + (\rho p - M\dot{\rho}q)^2\right] \qquad (3.15)$$

Or using equations 3.3,

$$I(t) = \frac{1}{2}\left[\frac{q^2}{\rho^2} + M^2(\rho\dot{q} - \dot{\rho}q)^2\right] \qquad (3.16)$$

The equation for σ will become,

$$M^2\ddot{\rho} + \Omega^2\rho - \frac{1}{\rho^3} = 0 \qquad (3.17)$$

Equation 3.16 along with equation 3.17 defines the class of invariants for the Time Dependent Harmonic Oscillators known as the **Lewis-Ermakov Invariants**.

4 Solution For $\rho(t)$ In The Case Of Constant Frequency Ω

We start with equation 3.17. It has one trivial solution,

$$\rho(t) = \pm|\Omega|^{-\frac{1}{2}} \qquad (4.1)$$

But it is possible to obtain more general solutions to 3.17 than this solution. Multiply 3.17 by $\dot{\rho}$ and then the first integral is easily obtained.

$$
\begin{gathered}
\frac{M^2}{2}\int\frac{\mathrm{d}(\dot{\rho})^2}{\mathrm{d}t}\mathrm{d}t + \Omega^2\int\rho\mathrm{d}\rho - \int\rho^{-3}\mathrm{d}\rho = K \\
M^2\dot{\rho}^2 + \Omega^2\rho^2 + \frac{1}{\rho^2} = 2K
\end{gathered}
$$

For ρ to have a real solution, the right hand side must be greater than or equal to $2|\Omega|$. Hence,

$$M^2\dot{\rho}^2 + \Omega^2\rho^2 + \frac{1}{\rho^2} = 2|\Omega|\cosh\delta \qquad \textit{where} \qquad \delta\in\mathbb{R} \qquad (4.2)$$

From this,

$$\dot{\rho} = \pm\frac{1}{M\rho}\sqrt{(2|\Omega|\cosh\delta)\rho^2 - \Omega^2\rho^4 - 1}$$

$$\Rightarrow \qquad \frac{\rho \mathrm{d}\rho}{\sqrt{\left(\frac{2}{|\Omega|}\cosh\delta\right)\rho^2 - \rho^4 - \frac{1}{\Omega^2}}} = \pm\frac{|\Omega|}{M}\mathrm{d}t$$

Substitute $u = \rho^2$. Then,

$$\frac{1}{2}\frac{\mathrm{d}u}{\sqrt{\left(\frac{2}{|\Omega|}\cosh\delta\right)u - u^2 - \frac{1}{\Omega^2}}} = \pm\frac{|\Omega|}{M}\mathrm{d}t$$

Now,

$$\begin{aligned}\left(\frac{2}{|\Omega|}\cosh\delta\right)u - u^2 - \frac{1}{\Omega^2} &= -\left(u - \frac{1}{|\Omega|}\cosh\delta\right)^2 + \frac{\cosh^2\delta - 1}{|\Omega|^2} \\ &= -\left(u - \frac{1}{|\Omega|}\cosh\delta\right)^2 + \frac{\sinh^2\delta}{|\Omega|^2}\end{aligned}$$

Put $z = u - \frac{1}{|\Omega|}\cosh\delta$. Then the integral will become,

$$\begin{aligned}\frac{1}{2}\cdot\frac{|\Omega|}{\sinh\delta}\int^z \frac{\mathrm{d}z'}{\sqrt{1 - \left(\frac{z'|\Omega|}{\sinh\delta}\right)^2}} &= \pm\frac{\Omega}{M}\int^t \mathrm{d}t' \\ \arcsin\left(\frac{z|\Omega|}{\sinh\delta}\right)\cdot\frac{\sinh\delta}{|\Omega|} &= \pm\frac{2\sinh\delta}{M}t + K' \\ \Rightarrow \qquad z &= \pm\frac{\sinh\delta}{|\Omega|}\sin\left(\frac{2\Omega}{M}t + \varphi\right) \\ \Rightarrow \qquad u &= \frac{1}{|\Omega|}\left[\cosh\delta \pm \sinh\delta\sin\left(\frac{2\Omega}{M}t + \varphi\right)\right]\end{aligned}$$

Hence,

$$\rho(t) = \pm|\Omega|^{-\frac{1}{2}}\left[\cosh\delta \pm \sinh\delta\sin\left(\frac{2\Omega}{M}t + \varphi\right)\right]^{\frac{1}{2}} \tag{4.3}$$

Where $\varphi \in \mathbb{R}$ is the constant of integration.
$\delta = 0$ will yield the trivial solution that we had in the beginning of the section.

5 Quantisation of The Lewis-Ermakov Invariant

$$I(t) = \frac{1}{2}\left[\frac{q^2}{\rho^2} + (\rho p - M\dot{\rho}q)^2\right]$$

We would like to write the invariant in the form,

$$I(t) = \hbar(a^\dagger a + \frac{1}{2}) \tag{5.1}$$

For that, from the analogy of Time Independent Harmonic Oscillator, define,

$$
\begin{aligned}
a &= \frac{1}{\sqrt{2\hbar}}\left(\frac{q}{\rho} + i(\rho p - M\dot{\rho}q)\right)\\
a^\dagger &= \frac{1}{\sqrt{2\hbar}}\left(\frac{q}{\rho} - i(\rho p - M\dot{\rho}q)\right)
\end{aligned}
\tag{5.2}
$$

Then,

$$
\begin{aligned}
a^\dagger a &= \frac{1}{2\hbar}\left[\frac{q^2}{\rho^2} + (\rho p - M\dot{\rho}q)^2 + \frac{i}{\rho}[q,(\rho p - M\dot{\rho}q)]\right]\\
&= \frac{1}{2\hbar}[2I - \hbar]
\end{aligned}
$$

Which on rearrangement will give 5.1. Define $N = a^\dagger a$. N and I will have common eigenstates as these operators commute. Then, let $|s\rangle$ be the eigenstate of N with eigenvalue s. i.e,

$$N|s\rangle = s|s\rangle \tag{5.3}$$

The commutation relations that defines the algebra are these,

$$
\begin{aligned}
[a, a^\dagger] &= 1\\
[N, a^\dagger] &= a^\dagger\\
[N, a] &= -a
\end{aligned}
\tag{5.4}
$$

a and $a^\dagger$ have the same properties as that of the creation and annihilation operators of the Time Independent Harmonic Oscillator. The underlying algebra is the same. Hence,

$$
\begin{aligned}
a|s\rangle &= \sqrt{s}\;|s-1\rangle\\
a^\dagger|s\rangle &= \sqrt{s+1}\;|s+1\rangle
\end{aligned}
\tag{5.5}
$$

Demand,

$$a|0\rangle = 0 \tag{5.6}$$

Without this demand, the expectation value of energy(which will be found soon) will have negative values. With this demand, the eigenvalues of $I(t)$ denoted by λ_s is obtained as,

$$\lambda_s = \hbar(s + \frac{1}{2}) \tag{5.7}$$

To get the eigenstates of I which solves the Schrodinger equation, we need to find the phase given by equation 1.9. For that write,

$$
\begin{aligned}
q &= \sqrt{\frac{\hbar}{2}}\,\rho(a^\dagger + a)\\
p &= \sqrt{\frac{\hbar}{2}}\left[a\left(M\dot{\rho} - \frac{i}{\rho}\right) + a^\dagger\left(M\dot{\rho} + \frac{i}{\rho}\right)\right]
\end{aligned}
\tag{5.8}
$$

Then using this,

$$H(t) = \frac{\hbar}{4M}\left\{ \left[\Omega^2\rho^2 + \left(M\dot{\rho} - \frac{i}{\rho}\right)^2\right] a^2 + \left[\Omega^2\rho^2 + \left(M\dot{\rho} + \frac{i}{\rho}\right)^2\right] (a^\dagger)^2 \right.$$
$$\left. + \left(M^2\dot{\rho}^2 + \Omega^2\rho^2 + \frac{1}{\rho^2}\right)\left\{a, a^\dagger\right\} \right\} \qquad (5.9)$$

But $\langle s| H |s\rangle$ will have contribution only from the $\left\{a, a^\dagger\right\}$ term! Then,

$$\begin{aligned}\langle s| H |s\rangle &= \frac{\hbar}{4M}\left(M^2\dot{\rho}^2 + \Omega^2\rho^2 + \frac{1}{\rho^2}\right)\langle s| \left\{a, a^\dagger\right\} |s\rangle \\ &= \frac{\hbar}{4M}\left(M^2\dot{\rho}^2 + \Omega^2\rho^2 + \frac{1}{\rho^2}\right)\langle s| (2N+1) |s\rangle\end{aligned}$$

$$\Rightarrow \qquad \langle s| H |s\rangle = \frac{1}{2M}\left(M^2\dot{\rho}^2 + \Omega^2\rho^2 + \frac{1}{\rho^2}\right)(s + \frac{1}{2})\hbar \qquad (5.10)$$

Note that if s did not have a minimum value of zero, the expectation value of energy will have negative values unbounded below.
To find the phase, we need $\langle s| \frac{\partial}{\partial t} |s\rangle$ also. For that, start with,

$$a^\dagger |s-1\rangle = \sqrt{s}\, |s\rangle$$

Differentiate with time and then multiply by $\langle s|$ from the left. Then after rearranging,

$$\langle s| \frac{\partial}{\partial t} |s\rangle = \langle s-1| \frac{\partial}{\partial t} |s-1\rangle + s^{-\frac{1}{2}} \langle s| \frac{\partial a^\dagger}{\partial t} |s-1\rangle$$

Now,

$$\frac{\partial a^\dagger}{\partial t} = \frac{1}{\sqrt{2\hbar}}\left[-\frac{\dot{\rho}}{\rho^2}q^2 - i(\dot{\rho}p - M\ddot{\rho}q)\right]$$

Use equations for q and p in terms of a and $a^\dagger$ to simplify this and get,

$$\frac{\partial a^\dagger}{\partial t} = \frac{1}{2}\left[\left(-\frac{2\dot{\rho}}{\rho} + i(-M\dot{\rho}^2 + M\ddot{\rho}\rho)\right) a + i(M\ddot{\rho}\rho - M\dot{\rho}^2)a^\dagger\right] \qquad (5.11)$$

Using this,

$$\begin{aligned}\langle s| \frac{\partial}{\partial t} |s\rangle &= \langle s-1| \frac{\partial}{\partial t} |s-1\rangle + i\frac{M}{2}(\rho\ddot{\rho} - \dot{\rho}^2) \\ &= \langle s-2| \frac{\partial}{\partial t} |s-2\rangle + 2i\frac{M}{2}(\rho\ddot{\rho} - \dot{\rho}^2) \\ &\vdots \\ &= \langle 0| \frac{\partial}{\partial t} |0\rangle + i\frac{s}{2}M(\rho\ddot{\rho} - \dot{\rho}^2)\end{aligned}$$

$\langle 0| \frac{\partial}{\partial t} |0\rangle$ cannot be found from this and it is arbitrary. Without loss of generality, choose,

$$\langle 0| \frac{\partial}{\partial t} |0\rangle = i\frac{M}{4}(\rho\ddot{\rho} - \dot{\rho}^2)$$

Then,

$$\langle s| \frac{\partial}{\partial t} |s\rangle = i\frac{M}{2}(\rho\ddot{\rho} - \dot{\rho}^2)(s + \frac{1}{2}) \tag{5.12}$$

Then using 1.9,

$$\begin{aligned} \hbar\frac{\mathrm{d}\alpha_s}{\mathrm{d}t} &= -\frac{\hbar}{2}M(\rho\ddot{\rho} - \dot{\rho}^2)(s + \frac{1}{2}) - \frac{\hbar}{2M}(M^2\dot{\rho}^2 + \Omega^2\rho^2 + \rho^{-2})(s + \frac{1}{2}) \\ &= \frac{\hbar}{2M}(s + \frac{1}{2})[-\rho(M^2\ddot{\rho} + \Omega^2\rho) - \rho^{-2}] \\ &= \frac{\hbar}{M}(s + \frac{1}{2})\rho^{-2} \end{aligned}$$

Where 3.17 is used in the second step. Hence,

$$\alpha_s(t) = -\frac{1}{M}(s + \frac{1}{2})\int^t \frac{1}{\rho^2(t')}\mathrm{d}t' \tag{5.13}$$

Hence the eigenstates of $I(t)$ given by,

$$|\Psi(t)\rangle = e^{i\alpha_s(t)} |s\rangle \tag{5.14}$$

Satisfy the Schrodinger equation!

6 Transition Probabilities

Assume that the hamiltonian in the remote past was a constant, with a frequency $\Omega_1 > 0$.

$$H(-\infty) = \frac{1}{2M}[p^2 + \Omega_1^2 q^2] \tag{6.1}$$

Using 4.3 choose a particular solution for ρ corresponding to $\delta = 0$. Hence,

$$\rho(-\infty) = \Omega_1^{-\frac{1}{2}} \tag{6.2}$$

Then,

$$\begin{aligned} I(-\infty) &= \frac{1}{2\Omega_1}[p^2 + \Omega_1^2 q^2] \\ &= \frac{M}{\Omega_1}H(-\infty) \end{aligned}$$

As $I(-\infty)$ and $H(-\infty)$ commute, they share the a common set of eigenstates, denote by $|\lambda; -\infty\rangle$.
Now assume that the hamiltonian will settle into a constant in the distant future with a frequency $\Omega_2 > 0$. In general, $I(\infty)$ and $H(\infty)$ may not commute. They do so only if we choose a particular solution for ρ. Let $|m; f\rangle$ denote the eigenstates of $H(\infty)$ and $I(\infty)$ will

have eigenstates $|\lambda;\infty\rangle$.
Now assume the system starts off with the state $|n;i\rangle = |0;-\infty\rangle$. Then using 2.4,

$$T(0\to m) = \sum_\lambda e^{i(\alpha_\lambda(\infty)-\alpha_\lambda(-\infty))}\langle m;f|\lambda;\infty\rangle\langle\lambda;-\infty|n;i\rangle = e^{i(\alpha_0(\infty)-\alpha_0(-\infty))}\langle m;f|0;\infty\rangle \tag{6.3}$$

To find this, we use the following trick,
From the analysis of time independent harmonic oscillator, we know,

$$H(\infty) = \frac{1}{2M}[p^2+\Omega_2^2q^2] = \frac{\hbar\Omega_2}{M}(b^\dagger b+\frac{1}{2}) \tag{6.4}$$

Where,

$$\begin{aligned} b =& \sqrt{\frac{\Omega_2}{2\hbar}}\left(q+\frac{ip}{\Omega_2}\right) \\ b^\dagger =& \sqrt{\frac{\Omega_2}{2\hbar}}\left(q-\frac{ip}{\Omega_2}\right) \end{aligned} \tag{6.5}$$

And these operators obey the Heisenberg algebra. Using them,

$$\begin{aligned} q =& \sqrt{\frac{\hbar}{2\Omega_2}}(b^\dagger+b) \\ p =& i\sqrt{\frac{\hbar\Omega_2}{2}}(b^\dagger-b) \end{aligned} \tag{6.6}$$

We had defined ladder operators for $I(t)$ in 5.2. We can rewrite 5.2 using 6.5 as,

$$\begin{aligned} a(t) &= \frac{1}{\sqrt{2\hbar}}\left[\frac{1}{\rho}\sqrt{\frac{\hbar}{2\Omega_2}}(b^\dagger+b)+i\left(\rho i\sqrt{\frac{\hbar\Omega_2}{2}}(b^\dagger-b)-M\dot\rho\sqrt{\frac{\hbar}{2\Omega_2}}(b^\dagger+b)\right)\right] \\ &= \frac{1}{\sqrt{4\Omega_2}}\left[\frac{1}{\rho}+\rho\Omega_2-iM\dot\rho\right]b+\frac{1}{\sqrt{4\Omega_2}}\left[\frac{1}{\rho}-\rho\Omega_2-iM\dot\rho\right]b^\dagger \\ &= \eta(t)b+\zeta(t)b^\dagger \end{aligned}$$

Where,

$$\begin{aligned} \eta(t) &= \frac{1}{\sqrt{4\Omega_2}}\left[\frac{1}{\rho}+\rho\Omega_2-iM\dot\rho\right] \\ \zeta(t) &= \frac{1}{\sqrt{4\Omega_2}}\left[\frac{1}{\rho}-\rho\Omega_2-iM\dot\rho\right] \end{aligned} \tag{6.7}$$

We have,

$$\begin{aligned} a(t) &= \eta(t)b+\zeta(t)b^\dagger \\ a^\dagger(t) &= \zeta^*(t)b+\eta^*(t)b^\dagger \end{aligned} \tag{6.8}$$

Also,

$$\begin{aligned} [a,a^\dagger] &= |\eta|^2-|\zeta|^2 \\ &= 1 \end{aligned} \tag{6.9}$$

The equations 6.8 along with the above condition is known as the **Bogoliubov Transformations**.

Now, write,

$$|0;\infty\rangle = \sum_n |n;f\rangle \langle n;f|0;\infty\rangle$$

Then,

$$\begin{aligned}
a(\infty)|0;\infty\rangle &= (\eta b + \zeta b^\dagger)\sum_n |n;f\rangle \langle n;f|0;\infty\rangle \\
0 &= \sum_{n=1}^{\infty} \eta\sqrt{n}\,|n-1;f\rangle \langle n;f|0;\infty\rangle + \sum_{n=0}^{\infty} \zeta\sqrt{n+1}\,|n+1;f\rangle \langle n;f|0;\infty\rangle \\
&= \eta\,|0;f\rangle \langle 1;f|0;\infty\rangle + \sum_{n=1}^{\infty} |n;f\rangle \left[\eta\sqrt{n+1}\,\langle n+1;f|0;\infty\rangle + \zeta\sqrt{n}\,\langle n-1;f|0;\infty\rangle\right]
\end{aligned}$$

As $\{|n;f\rangle\}$ form a complete orthonormal basis, for the above equation to be true, we need each coefficient go to zero. Then,

$$\begin{aligned}
\langle 1;f|0;\infty\rangle &= 0 \\
\eta\sqrt{n+1}\,\langle n+1;f|0;\infty\rangle + \zeta\sqrt{n}\,\langle n-1;f|0;\infty\rangle &= 0
\end{aligned} \tag{6.10}$$

From the second of the above,

$$\langle n+1;f|0;\infty\rangle = -\frac{\zeta}{\eta}\left(\frac{n}{n+1}\right)^{\frac{1}{2}} \langle n-1;f|0;\infty\rangle$$

This recursion relation can be easily solved. As the $2r+1$ th constant is related to $c_1 = \langle 1;f|0;\infty\rangle = 0$, $\langle 2r+1;f|0;\infty\rangle = 0$. But,

$$\langle 2r;f|0;\infty\rangle = \left(-\frac{\zeta}{\eta}\right)^r \left(\frac{1.3.\cdots(2r-1)}{2.4.\cdots(2r)}\right)^{\frac{1}{2}} \langle 0;f|0;\infty\rangle = \left(-\frac{\zeta}{\eta}\right)^r \left(\frac{(2r)!}{2^{2r}(r!)^2}\right)^{\frac{1}{2}} \langle 0;f|0;\infty\rangle$$

Hence,

$$|0;\infty\rangle = \sum_{r=0}^{\infty} |2r;f\rangle \left(-\frac{\zeta}{\eta}\right)^r \left(\frac{(2r)!}{2^{2r}(r!)^2}\right)^{\frac{1}{2}} \langle 0;f|0;\infty\rangle \tag{6.11}$$

But using $\langle 0;\infty|0;\infty\rangle = 1$,

$$|\langle 0;f|0;\infty\rangle|^2 \sum_{r=0}^{\infty} \left|\frac{\zeta}{\eta}\right|^{2r} \frac{(2r)!}{2^{2r}(r!)^2} = 1$$

Which gives,

$$\begin{aligned}
|\langle 0;f|0;\infty\rangle| &= \left[\sum_{r=0}^{\infty} \left|\frac{\zeta}{\eta}\right|^{2r} \frac{(2r)!}{2^{2r}(r!)^2}\right]^{-1} \\
&= \left[\left(1-\left|\frac{\zeta}{\eta}\right|\right)^{-\frac{1}{2}}\right]^{-1} \\
&= \frac{1}{|\eta|}
\end{aligned}$$

Hence,

$$\begin{aligned}|T(0\to 0)| &= |\eta|^{-\frac{1}{2}}\\ &= (4\Omega_2)^{\frac{1}{4}}(\rho^{-2}+\Omega_2^2\rho^2+2\Omega_2+M^2\dot{\rho}^2)^{-\frac{1}{4}}\\ &= (4\Omega_2)^{\frac{1}{4}}(2\Omega_2\cosh\delta+2\Omega_2)^{-\frac{1}{4}}\\ &= \left(\frac{2}{1+\cosh\delta}\right)^{\frac{1}{4}}\end{aligned}$$

Hence,

$$P_{0\to 0}=\left(\frac{2}{1+\cosh\delta}\right)^{\frac{1}{2}} \tag{6.12}$$

Now,

$$\begin{aligned}P_{0\to 2m} &= |\langle 2m;f|0;\infty\rangle|^2\\ &= \left|\sum_{r=0}^{\infty}\langle 2m;f|2r;f\rangle\left(-\frac{\zeta}{\eta}\right)^r\left(\frac{(2r)!}{2^{2r}(r!)^2}\right)^{\frac{1}{2}}\left(\frac{2}{1+\cosh\delta}\right)^{\frac{1}{4}}\right|^2\\ &= \left|\frac{\zeta}{\eta}\right|^{2m}\frac{(2m)!}{2^{2m}(m!)^2}\left(\frac{2}{1+\cosh\delta}\right)^{\frac{1}{2}}\end{aligned}$$

But,

$$\left|\frac{\zeta}{\eta}\right|^2=\frac{\cosh\delta-1}{\cosh\delta+1}$$

Then,

$$P_{0\to 2m}=\frac{(2m)!}{2^{2m}(m!)^2}\left(\frac{\cosh\delta-1}{\cosh\delta+1}\right)^m\left(\frac{2}{1+\cosh\delta}\right)^{\frac{1}{2}}$$

Or in general,

$$P_{0\to m}=\begin{cases}0, & \text{if } m \text{ is odd.}\\ \frac{(m)!}{2^m((m/2)!)^2}\left(\frac{\cosh\delta-1}{\cosh\delta+1}\right)^{\frac{m}{2}}\left(\frac{2}{1+\cosh\delta}\right)^{\frac{1}{2}}, & \text{if } m \text{ is even.}\end{cases} \tag{6.13}$$

To find the general transition amplitude, Write,

$$\begin{aligned}|s;\infty\rangle &= \frac{(a^\dagger)^s}{\sqrt{s!}}|0;\infty\rangle\\ &= \frac{1}{\sqrt{s!}}\sum_n(\zeta^*(t)b+\eta^*(t)b^\dagger)^s|n;f\rangle\langle n;f|0;\infty\rangle\end{aligned}$$

Hence,

$$\begin{aligned}P_{s\to m} &= |\langle m;f|s;\infty\rangle|^2\\ &= \frac{1}{s!}\left(\frac{2}{1+\cosh\delta}\right)^{\frac{1}{2}}\left|\sum_r\left(-\frac{\zeta}{2\eta}\right)^r\frac{\sqrt{(2r)!}}{r!}\langle m;f|(\zeta^*(t)b+\eta^*(t)b^\dagger)^s|2r;f\rangle\right|^2\end{aligned} \tag{6.14}$$

7 Coherent States Of The Lewis Invariant[2]

The Lewis Invariant can be written in a factorized form as,

$$I(t) = \hbar(a^\dagger a + \frac{1}{2}) \tag{7.1}$$

Where a and $a^\dagger$ is defined by equation 5.2. In the last section, we saw that these operators can be obtained via a Bogoliubov transformation given by 6.8. The coherent states of $I(t)$ are the eigenstates of a. As a is obtained by the Bogoliubov transformations(Eq. 6.8), the coherent states of $I(t)$ are the squeezed states of the constant Hamiltonian, $H(\infty)$. The squeezing operator is defined as,

$$S(\epsilon) = e^{\frac{1}{2}\left[\epsilon^* b^2 - \epsilon (b^\dagger)^2\right]} \qquad where \qquad \epsilon = re^{i\theta} \tag{7.2}$$

We know,

$$\begin{aligned} S^\dagger b S &= b\cosh r - e^{i\theta} b^\dagger \sinh r \\ S b^\dagger S^\dagger &= -be^{-i\theta}\sinh r + b^\dagger \cosh r \end{aligned}$$

The above is also a Bogoliubov transformation. The squeezed states are obtained by acting S on $|0\rangle$ where $|0\rangle$ is the ground state of the time independent Hamiltonian. Hence,

$$|\epsilon\rangle = S(\epsilon)\,|0\rangle = e^{\frac{|\epsilon|^2}{4}} \sum_{n=0}^{\infty} \left(-\frac{\epsilon}{2}\right)^n \frac{\sqrt{(2n)!}}{n!}\,|2n\rangle \tag{7.3}$$

If we choose,

$$\epsilon = \frac{\zeta}{\eta} \tag{7.4}$$

Comparing with Eq. 6.11, the squeezed states of the constant hamiltonian are the same as coherent states of $I(t)$ apart from a normalization factor.

8 The Wronskian And The Lewis-Ermakov Invariant [4, 5]

The Lagrangian corresponding to the Hamiltonian given by equation 3.1 is,

$$L(t) = \frac{1}{2} M\dot{q}^2 - \frac{1}{2M}\Omega^2 q^2 \tag{8.1}$$

The Euler-Lagrange equations is straight forward, and obtained as,

$$M^2 \ddot{q}(t) + \Omega^2(t) q(t) = 0 \tag{8.2}$$

Take the Ansats that the solution $q(t)$ can be written in the amplitude phase form as,

$$q(t) = g(t)e^{i\xi(t)} \tag{8.3}$$

Where both $g(t)$ and $\xi(t)$ are real functions of time. Plug the ansats in the equation of motion to get,

$$\left[M^2(\ddot{g} - \dot{\xi}^2 g) + iM^2(2\dot{\xi}\dot{g} + \ddot{\xi} g) + \Omega^2 g\right] e^{i\xi} = 0 \tag{8.4}$$

Equating the real and imaginary part separately to zero,

$$
\begin{aligned}
M^2\ddot{g} + \Omega^2 g - M^2\dot{\xi}^2 g &= 0 \\
2\dot{\xi}\dot{g} + \ddot{\xi} g &= 0
\end{aligned}
\tag{8.5}
$$

From the theory of differential Equations, we know that the Wronskian of the system is an invariant if the differential equation does not involve a first derivative of the dynamical variable (Abel's Theorem). If $q_1(t)$ and $q_2(t)$ are the linearly independent solutions of the equation of motion, then the Wronskian is defined by,

$$W = q_1(t)\dot{q}_2(t) - q_2(t)\dot{q}_1(t) \tag{8.6}$$

If $q_1(t) = g(t)e^{i\xi(t)}$ then $q_2(t) = g(t)e^{-i\xi(t)}$. Plug these in the equation of the Wronskian to get,

$$W = -2i\dot{\xi}g^2$$

As W is an invariant, we can scale it by any number. Hence take,

$$W = M\dot{\xi}g^2 \tag{8.7}$$

Use this relation to eliminate the $\dot{\xi}^2$ in the first of 8.5 and get,

$$M^2\ddot{g} + \Omega^2 g - \frac{W^2}{g^3} = 0 \tag{8.8}$$

This is exactly the equation we had for ρ equation3.17. In equation 3.17, $W = 1$. But as wronskian is an invariant, we can put it to unity by suitably scaling q_1 & q_2. Hence, the amplitude part of the solution for the equations of motion is the ρ! Now,

$$
\begin{aligned}
&& \dot{W} &= 0 \\
\Rightarrow && g(2\dot{\xi}\dot{g} + \ddot{\xi}g) &= 0
\end{aligned}
$$

Which is the second of 8.5.
Now, to find an equation for ξ, write,

$$g(t) = \sqrt{\frac{W}{M\dot{\xi}}} \tag{8.9}$$

Substitute this in the g equation. Rewrite $\dot{\xi} = \omega$. Then,

$$\omega\ddot{\omega} - \frac{3}{2}\dot{\omega}^2 + 2\omega^2(\omega^2 - \Omega^2) = 0$$

Hence, we have,

$$q(t) = \rho(t)e^{i\xi(t)} \tag{8.10}$$

$$M^2\ddot{\rho} + \Omega^2\rho - \frac{W^2}{\rho^3} = 0 \tag{8.11}$$

$$\omega\ddot{\omega} - \frac{3}{2}\dot{\omega}^2 + 2\omega^2(\omega^2 - \Omega^2) = 0 \tag{8.12}$$

With $\omega = \dot{\xi}$.
Now consider the real parts of q_1 and q_2 the independent solutions. Then the amplitude function is obtained by,

$$q_1^2 + q_2^2 = \rho^2 \tag{8.13}$$

Write $q = q_1$ and $q_2 = \sqrt{\rho^2 - q^2}$. Then,

$$\dot{q}_2 = \frac{\rho\dot{\rho} - q\dot{q}}{\sqrt{\rho^2 - q^2}}$$

Then the Wronskian is obtained as,

$$W = q\frac{\rho\dot{\rho} - q\dot{q}}{\sqrt{\rho^2 - q^2}} - \sqrt{\rho^2 - q^2}\dot{q} = \frac{\rho}{\sqrt{\rho^2 - q^2}}(q\dot{\rho} - \rho\dot{q}) \tag{8.14}$$

Now,

$$\begin{aligned} W^2 &= \frac{\rho^2}{\rho^2 - q^2}(q\dot{\rho} - \rho\dot{q})^2 \\ &= \frac{\rho^2 - q^2 + q^2}{\rho^2 - q^2}(q\dot{\rho} - \rho\dot{q})^2 \\ &= (q\dot{\rho} - \rho\dot{q})^2 + W^2\frac{q^2}{\rho^2} \end{aligned}$$

From this it is easy to see that the Lewis - Ermakov Invariants defined by equation 3.16 is nothing but,

$$I(t) = \frac{1}{2}W^2 = \frac{1}{2}\left[W^2\frac{q^2}{\rho^2} + (q\dot{\rho} - \rho\dot{q})^2\right] \tag{8.15}$$

In the definition 3.16, $W = 1$. But this can always be achieved by a scaling of the independent solutions q_1 and q_2. The mass term we had in 3.16 is also absent here. But it is not a problem. We can multiply the whole equation by M^2 and redefine $I = M^2 I$ and $W = MW$ to retain the exact form of the Lewis - Ermakov Invariant.

9 An Exactly Solvable Class of Problems In TDHO [6]

Consider a harmonic oscillator problem with both time dependent mass and time dependent frequency.

$$H(t) = \frac{p^2}{2M(t)} + \frac{1}{2}M(t)\omega^2(t)q^2 \tag{9.1}$$

Then,

$$\begin{aligned} \dot{q} &= \frac{p}{M(t)} \\ \dot{p} &= -M(t)\omega^2(t)q \end{aligned} \tag{9.2}$$

By differentiating the q equation and substitution from the other yield,

$$\ddot{q} + \frac{\dot{M}(t)}{M(t)}\dot{q} + \omega^2(t)q = 0 \tag{9.3}$$

At time $t = 0$, let $A_1(t)$ and $A_2(t)$ be two linearly independent solutions. i.e, $A_1(0) = \dot{A}_2(0) = 1$ and $\dot{A}_1(0) = A_2(0) = 0$. Then any general solution $q(t)$ can be written as,

$$q(t) = A_1(t)q(0) + A_2(t)\dot{q}(0) \tag{9.4}$$

If $q_1(t)$ and $q_2(t)$ are two linearly independent solutions of 9.3, then

$$\begin{aligned} A_1(t) &= \frac{q_1(t)\dot{q}_2(0) - q_2(t)\dot{q}_1(0)}{q_1(0)\dot{q}_2(0) - q_2(0)\dot{q}_1(0)} \\ A_2(t) &= \frac{q_2(t)q_1(0) - q_1(t)q_2(0)}{q_1(0)\dot{q}_2(0) - q_2(0)\dot{q}_1(0)} \end{aligned} \tag{9.5}$$

As the dynamical equation of the system 9.3 has a $\dot{q}$ part, the Wronskian of the system is not an invariant. By Abel's Theorem,

$$W(t) = W(0)e^{-\int \frac{\dot{M}(t)}{M(t)}\mathrm{d}t} = \frac{W(0)}{M(t)} \tag{9.6}$$

Define, a scaled Wronskian, W_s as,

$$W_s = W(t)M(t) \tag{9.7}$$

Obviously, W_s is time independent. From the general solution we have wrote down for 9.3, we can write the quantum mechanical counterpart as,

$$\begin{aligned} \hat{q}(t) &= A_1(t)\hat{q}(0) + A_2(t)\frac{\hat{p}(0)}{M(0)} \\ \hat{p}(t) &= M(t)\dot{A}_1(t)\hat{q}(0) + \dot{A}_2(t)M(t)\frac{\hat{p}(0)}{M(0)} \end{aligned} \tag{9.8}$$

Assume,

$$[\hat{q}(0), \hat{p}(0)] = i\hbar \tag{9.9}$$

If this relation holds for all times, then the quantisation of the system is done!

$$\begin{aligned} [\hat{q}(t), \hat{p}(t)] &= \frac{M(t)}{M(0)}\left(A_1(t)\dot{A}_2(t) - A_2(t)\dot{A}_1(t)\right)[\hat{q}(0), \hat{p}(0)] \\ &= \frac{M(t)}{M(0)}\frac{q_1(t)\dot{q}_2(t) - q_2(t)\dot{q}_1(t)}{q_1(0)\dot{q}_2(0) - q_2(0)\dot{q}_1(0)}\; i\hbar \\ &= \frac{W_s}{W_s}\; i\hbar = i\hbar \end{aligned}$$

The quantisation is done by defining the Ladder Operators,

$$\begin{aligned} a(t) &= \sqrt{\frac{M(t)\omega(t)}{2\hbar}}\left(\hat{q}(t) + \frac{i\hat{p}(t)}{M(t)\omega(t)}\right) \\ a^\dagger(t) &= \sqrt{\frac{M(t)\omega(t)}{2\hbar}}\left(\hat{q}(t) - \frac{i\hat{p}(t)}{M(t)\omega(t)}\right) \end{aligned} \tag{9.10}$$

Here,

$$[a(t), a^\dagger(t)] = 1 \qquad \forall t \tag{9.11}$$

And hence,

$$H(t) = \hbar\omega(t)\left(a^\dagger(t)a(t) + \frac{1}{2}\right) \tag{9.12}$$

For obtaining a class of exactly solvable systems, We take the assumption that although mass and frequency are time dependent, their product is time independent. i.e.

$$M(t)\omega(t) = constant \tag{9.13}$$

differentiation of the above w.r.t. time gives,

$$\frac{\dot{M}(t)}{M(t)} = -\frac{\dot{\omega}(t)}{\omega(t)} \tag{9.14}$$

Hence, 9.3 become,

$$\ddot{q} - \frac{\dot{\omega}(t)}{\omega(t)}\dot{q} + \omega^2(t)q = 0 \tag{9.15}$$

The exact solutions to this problem is easily obtained. The two independent solutions are given by,

$$q_1(t) = e^{i\int^t \omega(t')\mathrm{d}t'} \qquad\qquad q_2(t) = e^{-i\int^t \omega(t')\mathrm{d}t'} \tag{9.16}$$

10 Coupled Harmonic Oscillator Systems

10.1 Time Independent Case[7]

Consider a coupled harmonic oscillator, with the Hamiltonian,

$$H = \frac{1}{2}[p_1^2 + p_2^2 + k_0(x_1^2 + x_2^2) + k_1(x_1 - x_2)^2] \tag{10.1}$$

We have assumed $m = 1$ and $\hbar = 1$.
To solve the problem, it is easy to work with the normal coordinates defined by,

$$x_\pm = \frac{x_1 \pm x_2}{\sqrt{2}} \tag{10.2}$$

Then the hamiltonian will take the form,

$$H = \frac{1}{2}\left[\left(p_+^2 + k_0x_+^2\right) + \left(p_-^2 + (k_0 + 2k_1)x_-^2\right)\right] \tag{10.3}$$

The problem is uncoupled in the normal coordinates. So the ground state wave functions is the product of each individual wave functions. i.e.

$$\psi_0(x_+, x_-) = \left(\frac{\omega_+\omega_-}{\pi^2}\right)^{\frac{1}{4}} e^{-\frac{1}{2}(\omega_+x_+^2 + \omega_-x_-^2)} \tag{10.4}$$

Where,

$$\omega_+ = \sqrt{k_0} \qquad \omega_- = \sqrt{k_0 + 2k_1} \tag{10.5}$$

We can rewrite the wavefunction in terms of the old variables as,

$$\psi_0(x_1, x_2) = \left(\frac{\omega_+\omega_-}{\pi^2}\right)^{\frac{1}{4}} \exp\left\{-\frac{1}{4}[(\omega_+ + \omega_-)(x_1^2 + x_2^2) - 2(\omega_+ - \omega_-)x_1x_2]\right\} \tag{10.6}$$

Then the density matrix of an ensemble which is purely in the ground state is given by,

$$\rho(x_1, x_2, x_1', x_2') = \psi_0^*(x_1', x_2')\psi_0(x_1, x_2) \tag{10.7}$$

To get the density matrix for one oscillator alone, we find the trace of the full density matrix along one oscillator. The,

$$\begin{aligned}\rho_{out}(x_2, x_2') &= \int_{-\infty}^{\infty} \mathrm{d}x_1\, \rho(x_1, x_2, x_1, x_2') \\ &= \sqrt{\frac{\omega_+\omega_-}{\pi^2}} e^{-\frac{1}{4}(\omega_++\omega_-)(x_2^2+x_2'^2)} \int_{-\infty}^{\infty} \mathrm{d}x_1\, e^{-\frac{1}{2}[(\omega_++\omega_-)x_1^2+(\omega_+-\omega_-)(x_2+x_2')x_1]} \\ &= \sqrt{\frac{2\omega_+\omega_-}{\pi(\omega_++\omega_-)}} \exp\left\{-\frac{1}{4}(\omega_++\omega_-)(x_2^2+x_2'^2) + \frac{(\omega_+-\omega_-)^2(x_2+x_2')^2}{8(\omega_++\omega_-)}\right\}\end{aligned}$$

Define $\beta = \frac{(\omega_+-\omega_-)^2}{4(\omega_++\omega_-)}$ and $\beta - \frac{(\omega_++\omega_-)}{2} = -\gamma$. Then $\gamma - \beta = \frac{2\omega_+\omega_-}{(\omega_++\omega_-)}$. Hence,

$$\rho_{out}(x_2, x_2') = \sqrt{\frac{\gamma-\beta}{\pi}} \exp\left\{-\frac{\gamma}{2}(x_2^2+x_2'^2) + \beta x_2 x_2'\right\} \tag{10.8}$$

Now,

$$\begin{aligned}Tr(\rho_{out}) &= \sqrt{\frac{\gamma-\beta}{\pi}} \int_{-\infty}^{\infty} \mathrm{d}x_2\, e^{-(\gamma-\beta)x_2^2} \\ &= 1\end{aligned}$$

$$\begin{aligned}Tr(\rho_{out}^2) &= \frac{\gamma-\beta}{\pi} \int_{-\infty}^{\infty} \mathrm{d}x_2\, e^{-2(\gamma-\beta)x_2^2} \\ &= \sqrt{\frac{\gamma-\beta}{2\pi}} \\ &= \sqrt{\frac{\omega_+\omega_-}{\pi(\omega_++\omega_-)}}\end{aligned}$$

Now, the eigenvalue equation of the density matrix is,

$$\int_{-\infty}^{\infty} \mathrm{d}x'\, \rho_{out}(x, x') f_n(x') = p_n f_n(x) \tag{10.9}$$

The eigenvalues and eigenfunctions are known to be of the form,

$$f_n(x) = H_n(\sqrt{\alpha}x)e^{-\frac{\alpha x^2}{2}} \tag{10.10}$$

$$p_n = (1-\xi)\xi^n \tag{10.11}$$

Where $\alpha = \sqrt{\gamma^2 - \beta^2}$ and $\xi = \frac{\beta}{\alpha+\gamma}$.
Hence the entropy can be found using,

$$\begin{aligned}S &= -\sum_n p_n \log p_n \\ &= -(1-\xi)\log(1-\xi)\sum_n \xi^n - (1-\xi)\log\xi \sum_n n\xi^n \\ &= -\log(1-\xi) - \frac{\xi}{1-\xi}\log\xi \end{aligned} \tag{10.12}$$

10.2 Time Dependent Case

The Hamiltonian of the system is,

$$H(t) = \frac{1}{2}[p_1^2 + p_2^2 + \omega_0^2(x_1^2 + x_2^2) - \alpha(t)x_1x_2] \tag{10.13}$$

Where,

$$\alpha(t) = \begin{cases} \alpha_0, & \text{if } -\infty < t \leq 0 \\ 0, & \text{if } 0 < t < \infty \end{cases} \tag{10.14}$$

We are using $m = 1$ and $\hbar = 1$. In that case, the dimensions of physical quantities are as follows,

$$\begin{aligned} [E] &= T^{-1} \\ [L] &= T^{1/2} \\ [v] = [p] &= T^{-1/2} \\ [\omega] &= T^{-1} \\ [\alpha] &= T^{-2} \end{aligned}$$

Switch to normal coordinates, defined by $x_\pm = \frac{x_1 \pm x_2}{\sqrt{2}}$. Then,

$$H(t) = \frac{1}{2}\left[p_+^2 + \omega_+^2(t)x_+^2 + p_-^2 + \omega_-^2(t)x_-^2\right] \tag{10.15}$$

Where,

$$\omega_\pm^2(t) = \omega_0^2 \mp \frac{\alpha(t)}{2} \tag{10.16}$$

To make everything dimensionless, divide by ω_0. And define,

$$\begin{aligned} \tilde{H} &= \frac{H}{\omega_0} \\ \tilde{p}_\pm &= \frac{p_\pm}{\sqrt{\omega_0}} \\ \tilde{x}_\pm &= \sqrt{\omega_0}\, x_\pm \\ \tilde{\omega}_\pm &= \frac{\omega_\pm}{\omega_0} \\ \tilde{\alpha} &= \frac{\alpha}{\omega_0^2} \end{aligned}$$

Then, we have,

$$\tilde{H}(t) = \frac{1}{2}\left[\tilde{p}_+^2 + \tilde{\omega}_+^2(t)\tilde{x}_+^2 + \tilde{p}_-^2 + \tilde{\omega}_-^2(t)\tilde{x}_-^2\right] \tag{10.17}$$

Where,

$$\tilde{\omega}_\pm^2(t) = 1 \mp \frac{\tilde{\alpha}(t)}{2} \tag{10.18}$$

Then for $t \to -\infty$, the Hamiltonian is a constant and uncoupled in the normal coordinates. In the dimensionless variables,

$$\tilde{H}(-\infty) = \frac{1}{2}\left[\tilde{p}_+^2 + \tilde{\omega}_+^2\tilde{x}_+^2 + \tilde{p}_-^2 + \tilde{\omega}_-^2\tilde{x}_-^2\right] \tag{10.19}$$

Hence the ground state wave function of the system is given by,

$$\psi_0^{(-\infty)}(\tilde{x}_+, \tilde{x}_-) = \left(\frac{\tilde{\omega}_+\tilde{\omega}_-}{\pi^2}\right)^{\frac{1}{4}} \exp\left\{-\frac{1}{2}(\tilde{\omega}_+\tilde{x}_+^2 + \tilde{\omega}_-\tilde{x}_-^2)\right\} \tag{10.20}$$

At time $t = 0$, the Hamiltonian in the normal coordinates become,

$$\tilde{H}(0) = \frac{1}{2}\left[\tilde{p}_+^2 + \tilde{x}_+^2 + \tilde{p}_-^2 + \tilde{x}_-^2\right] \tag{10.21}$$

Then, any general normalized eigenstate is given by,

$$\psi_{mn}^{(0)}(\tilde{x}_+, \tilde{x}_-) = \frac{1}{\sqrt{2^{m+n}m!n!}}\left(\frac{1}{\pi}\right)^{\frac{1}{2}} \exp\left\{-\frac{1}{2}(\tilde{x}_+^2 + \tilde{x}_-^2)\right\} H_n(\tilde{x}_+)H_m(\tilde{x}_-) \tag{10.22}$$

Assume that the system was in the ground state at $t \to -\infty$. Then to understand the time evolution of the system after $t = 0$, we need to write the ground state as the linear combination of the eigenstates after time $t = 0$. i.e,

$$\psi_0^{(-\infty)}(\tilde{x}_+, \tilde{x}_-) = \sum_n \sum_m C_{mn}\psi_{mn}^{(0)}(\tilde{x}_+, \tilde{x}_-) \tag{10.23}$$

To find the coefficients,

$$\begin{aligned} C_{mn} &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \mathrm{d}\tilde{x}_+\mathrm{d}\tilde{x}_-\ \psi_0^{(-\infty)}(\tilde{x}_+, \tilde{x}_-)\left(\psi_{mn}^{(0)}(\tilde{x}_+, \tilde{x}_-)\right)^* \\ &= K_{mn}\left(\int_{-\infty}^{\infty} \mathrm{d}\tilde{x}_+\ e^{-\frac{\tilde{\omega}_+ + 1}{2}\tilde{x}_+^2} H_n(\tilde{x}_+)\right) \cdot \left(\int_{-\infty}^{\infty} \mathrm{d}\tilde{x}_-\ e^{-\frac{\tilde{\omega}_- + 1}{2}\tilde{x}_-^2} H_m(\tilde{x}_-)\right) \end{aligned} \tag{10.24}$$

Where,

$$K_{mn} = \frac{1}{\sqrt{2^{m+n}m!n!}}\left(\frac{1}{\pi}\right)^{\frac{1}{2}}\left(\frac{\tilde{\omega}_+\tilde{\omega}_-}{\pi^2}\right)^{\frac{1}{4}}$$

We need to evaluate the integral,

$$I_n = \int_{-\infty}^{\infty} \mathrm{d}x\ \exp\{-cx^2\}H_n(x)$$

Substitute $y = \sqrt{c}x$ and then,

$$I_n = \sqrt{\frac{1}{c}}\int \mathrm{d}y\ e^{-y^2} H_n\left(\right)$$

Using the formula from p. 504, equation 18 of *Integrals and Series Volume 2* by A.P. Prudnikov [3],

$$\int_{-\infty}^{\infty} \mathrm{d}x\ e^{-(x-y)^2} H_m(cx)H_n(cx) = \sqrt{\pi}\sum_{k=0}^{min(m,n)} 2^k k! \binom{m}{k}\binom{n}{k}\left(1 - c^2\right)^{\frac{m+n}{2} - k} H_{m+n-2k}\left(\frac{cy}{\sqrt{1 - c^2}}\right) \tag{10.25}$$

We get,

$$I_n = \sqrt{\frac{\pi}{c}}\left(1 - \frac{1}{c}\right)^{\frac{n}{2}} H_n(0)$$

But,

$$H_n(0) = \begin{cases} 0, & \text{if n is odd} \\ \frac{(-1)^{n/2} n!}{2^{n/2}(n/2)!}, & \text{if n is even} \end{cases}$$

Then using this result, we get $C_{mn} = 0$ when either n or m is odd, and,

$$C_{m,n} = K_{m,n} \frac{(-1)^{\frac{m+n}{2}}(m)!(n)!}{2^{\frac{m+n}{2}}(m/2)!(n/2)!} \frac{2\pi}{\sqrt{(\tilde{\omega}_+ + 1)(\tilde{\omega}_- + 1)}} \left(1 - \frac{2}{\tilde{\omega}_+ + 1}\right)^{\frac{n}{2}} \left(1 - \frac{2}{\tilde{\omega}_- + 1}\right)^{\frac{m}{2}} \tag{10.26}$$

When both m and n are even.
Hence,

$$\begin{aligned} \psi_0^{(-\infty)}(\tilde{x}_+, \tilde{x}_-) = \sum_{n\in E} \sum_{m\in E} \frac{(-1)^{\frac{m+n}{2}}}{2^{\frac{3(m+n)}{2}-1}(m/2)!(n/2)!} \frac{1}{\sqrt{(\tilde{\omega}_+ + 1)(\tilde{\omega}_- + 1)}} \left(\frac{\tilde{\omega}_+\tilde{\omega}_-}{\pi^2}\right)^{\frac{1}{4}} \\ \times \left(1 - \frac{2}{\tilde{\omega}_+ + 1}\right)^{\frac{n}{2}} \left(1 - \frac{2}{\tilde{\omega}_- + 1}\right)^{\frac{m}{2}} \exp\left\{-\frac{\tilde{x}_+^2 + \tilde{x}_-^2}{2}\right\} H_n(\tilde{x}_+) H_m(\tilde{x}_-) \end{aligned} \tag{10.27}$$

Where $E = \{0, 2, 4, \cdots\}$.
We can write this results in the actual original set of coordinates $\tilde{x}_1$ and $\tilde{x}_2$. Then the density matrix can be calculated as,

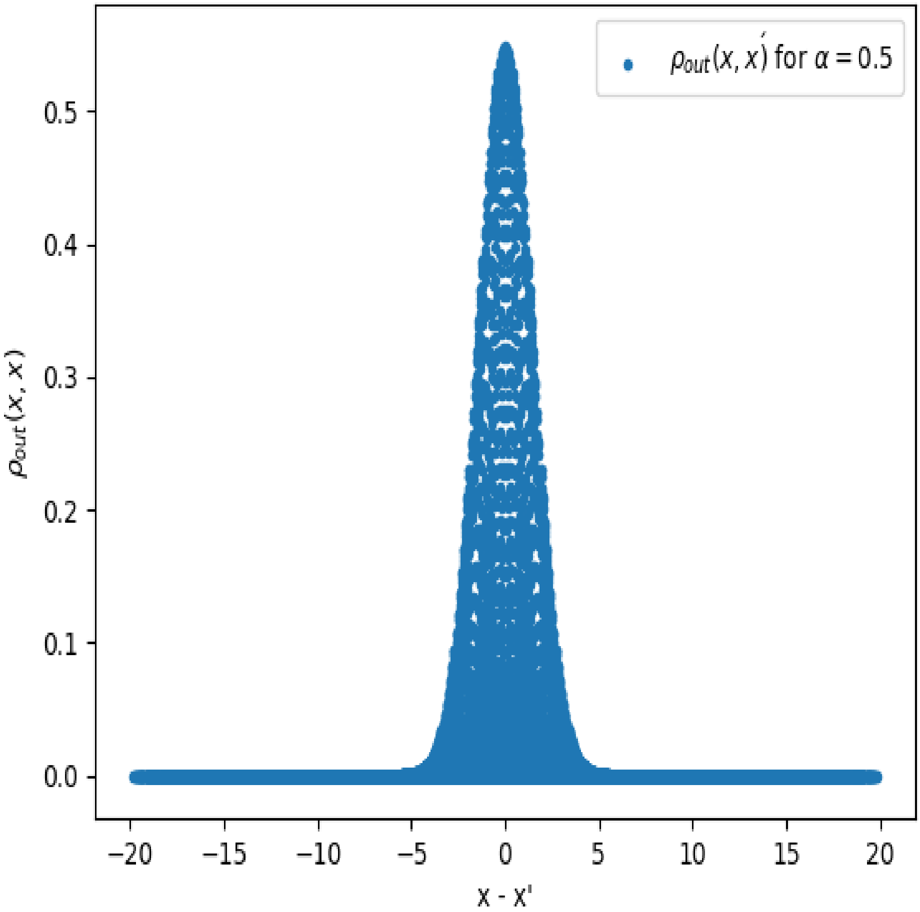
$$\rho(x_1, x_1', x_2, x_2') = \psi_0^{(-\infty)}(\tilde{x}_1, \tilde{x}_1)(\psi_0^{(-\infty)}(\tilde{x}_1', \tilde{x}_2'))^*$$

$$\begin{aligned} \rho(\tilde{x}_1, \tilde{x}_1', \tilde{x}_2, \tilde{x}_2') = \sum_{n\in E} \sum_{m\in E} \sum_{n'\in E} \sum_{m'\in E} A_{mn} A_{m'n'} H_n\left(\frac{\tilde{x}_1 + \tilde{x}_2}{\sqrt{2}}\right) H_m\left(\frac{\tilde{x}_1 - \tilde{x}_2}{\sqrt{2}}\right) \\ \times H_{n'}\left(\frac{\tilde{x}_1' + \tilde{x}_2'}{\sqrt{2}}\right) H_{m'}\left(\frac{\tilde{x}_1' - \tilde{x}_2'}{\sqrt{2}}\right) \end{aligned} \tag{10.28}$$

Then the trace is taken with respect to $\tilde{x}_2$ as,

$$\rho_{out}(\tilde{x}_1, \tilde{x}_1') = \int_{-\infty}^{\infty} \rho(\tilde{x}_1, \tilde{x}_1', \tilde{x}_2, \tilde{x}_2)\, d\tilde{x}_2 \tag{10.29}$$

This integral has been done using *Mathematica* The terms in the sum corresponding $n, n', m, m' \in \{0, 2\}$ contribute to ρ_{out}. These sixteen terms have been added up and plotted with respect to $x_1 - x_1'$ for coupling $\tilde{\alpha} = 0.5$.



10.3 Out Of Time Order Correlations(OTOC)

OTOC for one particle is defined to be[8],

$$C_{ij}(t) = -\langle n| [x_i(t), p_j(t_0)]^2 |n\rangle \tag{10.30}$$

Where t_0 is some initial time. For finding the OTOC for the coupled system, we analyse the system in the Heisenberg picture. After rescaling all the variables with ω_0, we have,

$$\tilde{H}(t) = \frac{1}{2}[\tilde{p}_+^2 + \tilde{p}_-^2 + \tilde{\omega}_+(\tilde{t})\tilde{x}_+^2 + \tilde{\omega}_-(\tilde{t})\tilde{x}_-^2]$$

Where,

$$\tilde{\omega}_\pm(\tilde{t}) = \begin{cases} \tilde{\omega}_\pm = 1 \mp \frac{\tilde{\alpha}}{2}, & \text{if } \tilde{t} < 0 \\ 1, & \text{if } \tilde{t} > 0 \end{cases} \tag{10.31}$$

and $\tilde{t} = \omega_0 t$. Now, start at an initial time, say $\tilde{t}_{-\infty} << 0$ such that,

$$[\tilde{x}_\pm(\tilde{t}_{-\infty}), \tilde{p}_\pm(\tilde{t}_{-\infty})] = i \tag{10.32}$$

Now, for any time $t < 0$,

$$\tilde{x}_\pm(\tilde{t}) = \tilde{x}_\pm(\tilde{t}_{-\infty}) \cos\left[\tilde{\omega}_\pm(\tilde{t} - \tilde{t}_{-\infty})\right] + \frac{\tilde{p}_\pm(\tilde{t}_{-\infty})}{\tilde{\omega}_\pm} \sin\left[\tilde{\omega}_\pm(\tilde{t} - \tilde{t}_{-\infty})\right] \tag{10.33}$$

$$\tilde{p}_\pm(\tilde{t}) = \tilde{p}_\pm(\tilde{t}_{-\infty}) \cos\left[\tilde{\omega}_\pm(\tilde{t} - \tilde{t}_{-\infty})\right] - \tilde{\omega}_\pm \tilde{x}_\pm(\tilde{t}_{-\infty}) \sin\left[\tilde{\omega}_\pm(\tilde{t} - \tilde{t}_{-\infty})\right] \tag{10.34}$$

We know that $\tilde{x}_1 = \frac{\tilde{x}_+ + \tilde{x}_-}{\sqrt{2}}$. Then for $\tilde{t} < 0$, the commutator,

$$[\tilde{x}_1(\tilde{t}), \tilde{p}_1(\tilde{t}_{-\infty})] = \frac{1}{2}[\tilde{x}_+(\tilde{t}) + \tilde{x}_-(\tilde{t}), \tilde{p}_+(\tilde{t}_{-\infty}) + \tilde{p}_-(\tilde{t}_{-\infty})]$$

As the normal coordinates are independent variables,

$$\begin{aligned}[\tilde{x}_1(\tilde{t}), \tilde{p}_1(\tilde{t}_{-\infty})] &= \frac{1}{2}\left\{[\tilde{x}_+(\tilde{t}), \tilde{p}_+(\tilde{t}_{-\infty})] + [\tilde{x}_-(\tilde{t}), \tilde{p}_-(\tilde{t}_{-\infty})]\right\} \\ &= \frac{i}{2}[\cos\left[\tilde{\omega}_+(\tilde{t} - \tilde{t}_{-\infty})\right] + \cos\left[\tilde{\omega}_-(\tilde{t} - \tilde{t}_{-\infty})\right]]\end{aligned}$$

Now, at $\tilde{t} = 0$,

$$\tilde{x}_\pm(0) = \tilde{x}_\pm(\tilde{t}_{-\infty}) \cos(\tilde{\omega}_\pm \tilde{t}_{-\infty}) - \frac{\tilde{p}_\pm(\tilde{t}_{-\infty})}{\tilde{\omega}_\pm} \sin(\tilde{\omega}_\pm \tilde{t}_{-\infty}) \tag{10.35}$$

$$\tilde{p}_\pm(0) = \tilde{p}_\pm(\tilde{t}_{-\infty}) \cos(\tilde{\omega}_\pm \tilde{t}_{-\infty}) + \tilde{\omega}_\pm \tilde{x}_\pm(\tilde{t}_{-\infty}) \sin(\tilde{\omega}_\pm \tilde{t}_{-\infty}) \tag{10.36}$$

These serves as the initial conditions for the next stage. The Hamiltonian after time $\tilde{t} = 0$ has frequancy 1 in the rescaled variables. Hence for any time $\tilde{t} > 0$,

$$\tilde{x}_\pm(\tilde{t}) = \tilde{x}_\pm(0) \cos \tilde{t} + \tilde{p}_\pm(0) \sin \tilde{t} \tag{10.37}$$

$$\tilde{p}_\pm(\tilde{t}) = \tilde{p}_\pm(0) \cos \tilde{t} - \tilde{x}_\pm(0) \sin \tilde{t} \tag{10.38}$$

Using this, the commutator,

$$[\tilde{x}_1(\tilde{t}), \tilde{p}_1(\tilde{t}_{-\infty})] = \frac{1}{2}[\tilde{x}_+(\tilde{t}) + \tilde{x}_-(\tilde{t}), \tilde{p}_+(\tilde{t}_{-\infty}) + \tilde{p}_-(\tilde{t}_{-\infty})]$$

$$\begin{aligned}[\tilde{x}_1(\tilde{t}), \tilde{p}_1(\tilde{t}_{-\infty})] &= \frac{1}{2}\left\{[\tilde{x}_+(\tilde{t}), \tilde{p}_+(\tilde{t}_{-\infty})] + [\tilde{x}_-(\tilde{t}), \tilde{p}_-(\tilde{t}_{-\infty})]\right\} \\ &= \frac{i}{2}\left\{\cos \tilde{t}\left[\cos(\tilde{\omega}_+ \tilde{t}_{-\infty}) + \cos(\tilde{\omega}_- \tilde{t}_{-\infty})\right] + \sin \tilde{t}\left[\tilde{\omega}_+ \sin(\tilde{\omega}_+ \tilde{t}_{-\infty}) + \tilde{\omega}_- \sin(\tilde{\omega}_- \tilde{t}_{-\infty})\right]\right\}\end{aligned}$$

Hence the OTOC can be easily calculated using Eq. 10.30,

$$C_{11}(\tilde{t}) = \begin{cases} \frac{1}{4}\left\{\cos \tilde{t}\left[\cos(\tilde{\omega}_+ \tilde{t}_{-\infty}) + \cos(\tilde{\omega}_- \tilde{t}_{-\infty})\right] + \sin \tilde{t}\left[\tilde{\omega}_+ \sin(\tilde{\omega}_+ \tilde{t}_{-\infty}) + \tilde{\omega}_- \sin(\tilde{\omega}_- \tilde{t}_{-\infty})\right]\right\}^2, & \text{if } \tilde{t} > 0 \\ \frac{1}{4}[\cos\left[\tilde{\omega}_+(\tilde{t} - \tilde{t}_{-\infty})\right] + \cos\left[\tilde{\omega}_-(\tilde{t} - \tilde{t}_{-\infty})\right]]^2 & \text{if } \tilde{t} < 0 \end{cases} \tag{10.39}$$

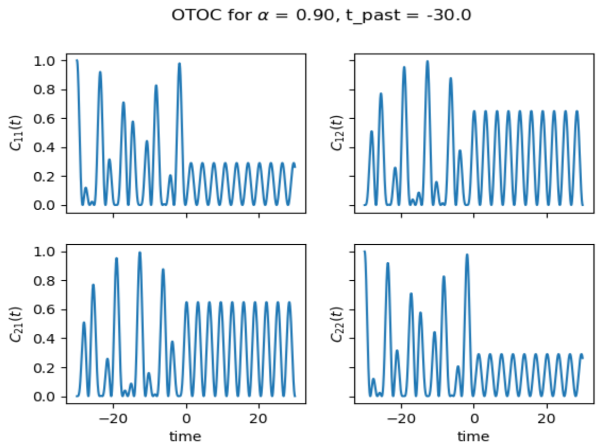
By the same method,

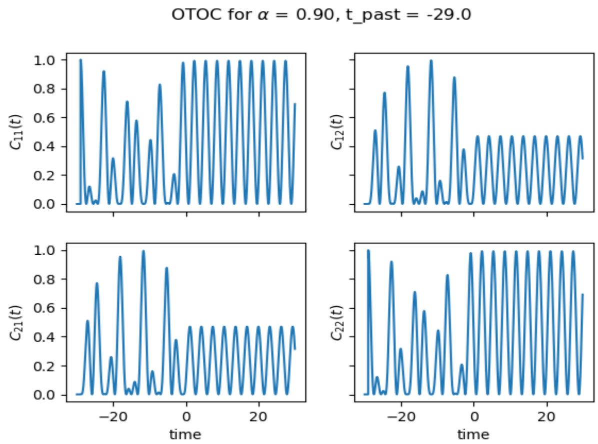
$$C_{12}(\tilde{t}) = \begin{cases} \frac{1}{4}\left\{\cos \tilde{t}\left[\cos(\tilde{\omega}_+ \tilde{t}_{-\infty}) - \cos(\tilde{\omega}_- \tilde{t}_{-\infty})\right] + \sin \tilde{t}\left[\tilde{\omega}_+ \sin(\tilde{\omega}_+ \tilde{t}_{-\infty}) - \tilde{\omega}_- \sin(\tilde{\omega}_- \tilde{t}_{-\infty})\right]\right\}^2, & \text{if } \tilde{t} > 0 \\ \frac{1}{4}[\cos\left[\tilde{\omega}_+(\tilde{t} - \tilde{t}_{-\infty})\right] - \cos\left[\tilde{\omega}_-(\tilde{t} - \tilde{t}_{-\infty})\right]]^2 & \text{if } \tilde{t} < 0 \end{cases} \tag{10.40}$$

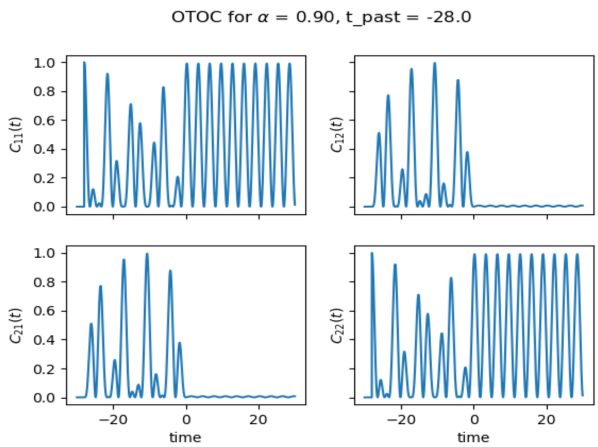
From the symmetry of the problem,

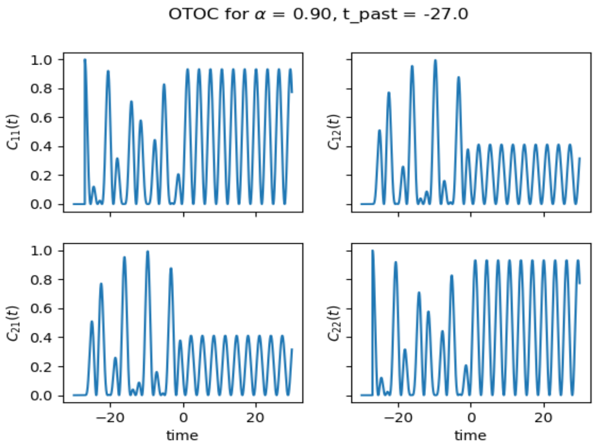
$$C_{22}(\tilde{t}) = C_{11}(\tilde{t}) \qquad C_{21}(\tilde{t}) = C_{12}(\tilde{t}) \tag{10.41}$$

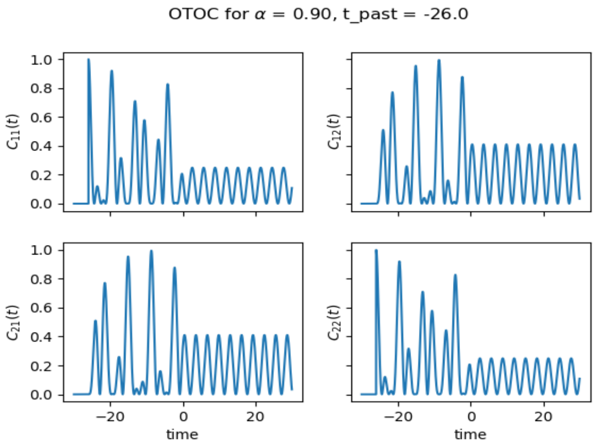
The OTOC functions are plotted for constant $\tilde{\alpha}$ but the starting time $\tilde{t}_{-\infty}$ is varied. Then we obtain,

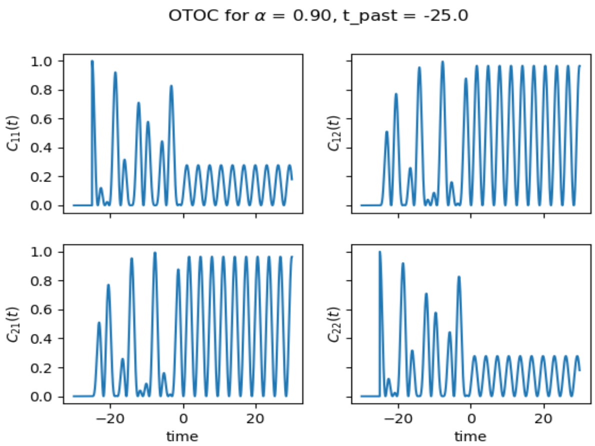




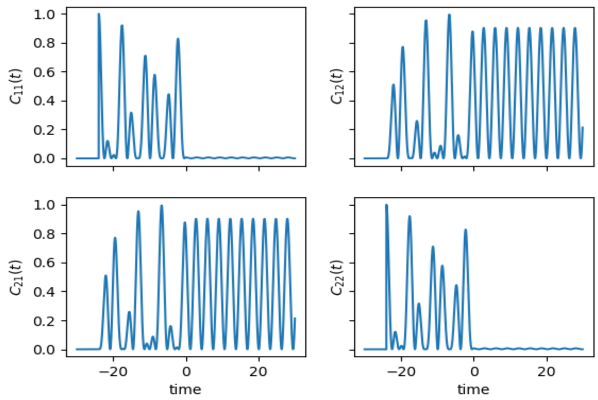



OTOC for α = 0.90, t_past = -27.0
$C_{11}(t)$
$C_{12}(t)$
$C_{21}(t)$
$C_{22}(t)$
time
time

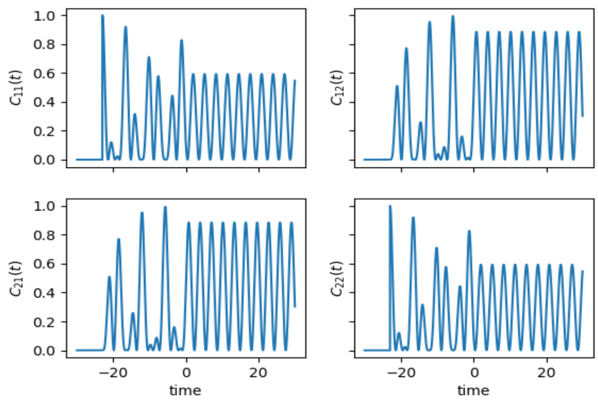

OTOC for α = 0.90, t_past = -26.0
$C_{11}(t)$
$C_{12}(t)$
$C_{21}(t)$
$C_{22}(t)$
time
time


OTOC for α = 0.90, t_past = -25.0
$C_{11}(t)$
$C_{12}(t)$
$C_{21}(t)$
$C_{22}(t)$
time
time

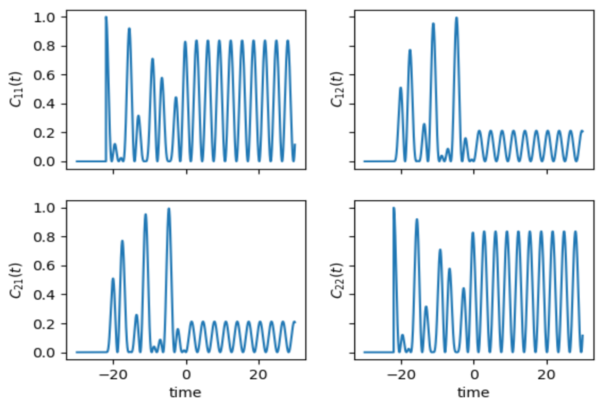
OTOC for α = 0.90, t_past = -24.0

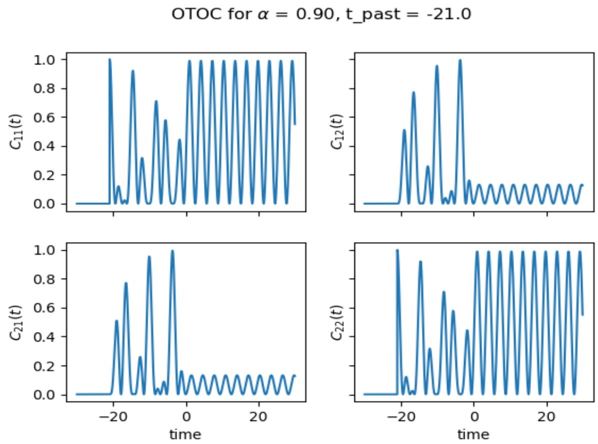


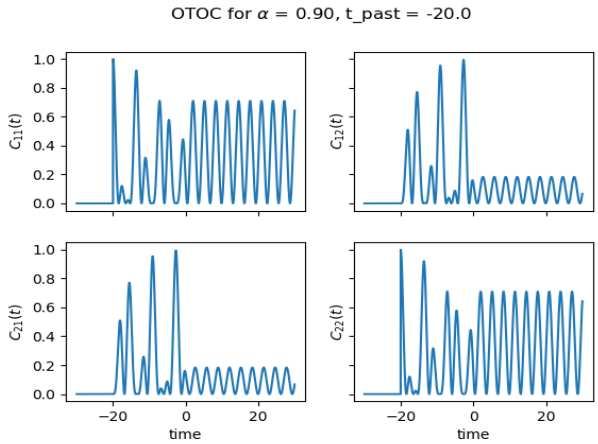
OTOC for α = 0.90, t_past = -23.0



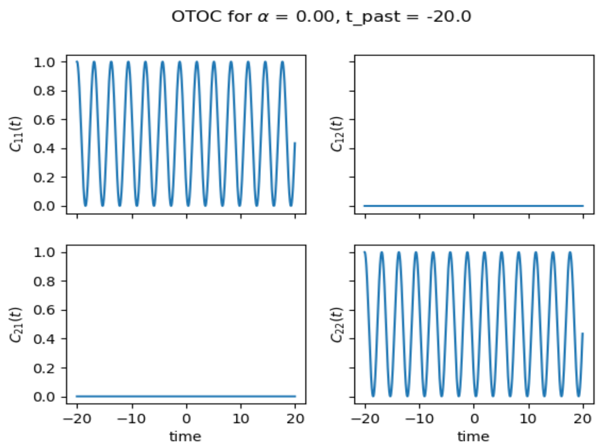
OTOC for α = 0.90, t_past = -22.0

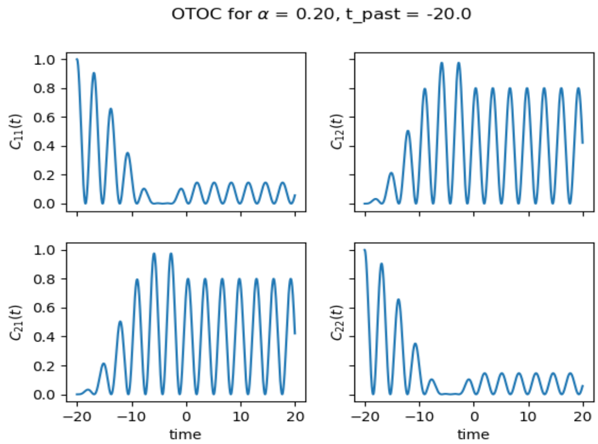


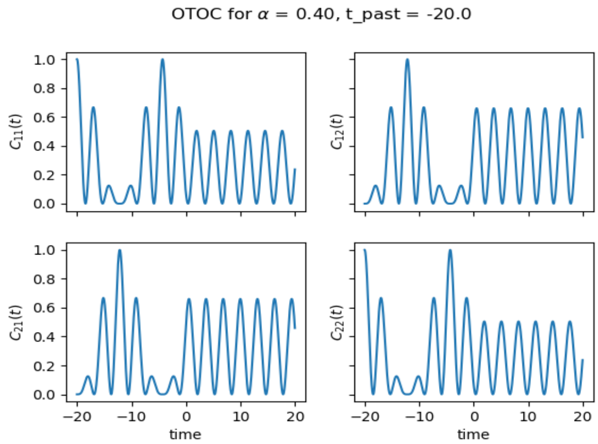


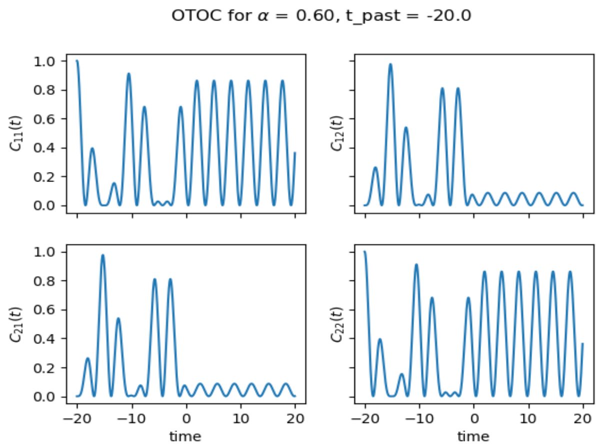


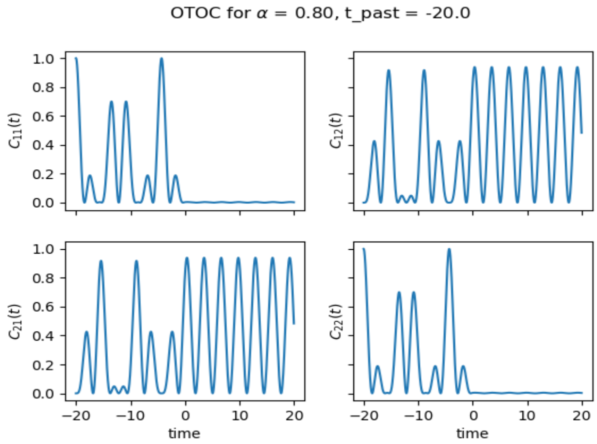
Also, the OTOC functions are plotted for constant $\tilde{t}_{-\infty}$ but varying $\tilde{\alpha}$. It can take values from -2.0 to 2.0. But the OTOC functions remain unchanged under the change $\tilde{\alpha} \to -\tilde{\alpha}$. Hence we plot only for positive values of $\tilde{\alpha}$.




OTOC for α = 0.20, t_past = -20.0
$C_{11}(t)$
$C_{12}(t)$
$C_{21}(t)$
$C_{22}(t)$
time
time


OTOC for α = 0.40, t_past = -20.0
$C_{11}(t)$
$C_{12}(t)$
$C_{21}(t)$
$C_{22}(t)$
time
time


OTOC for α = 0.60, t_past = -20.0
$C_{11}(t)$
$C_{12}(t)$
$C_{21}(t)$
$C_{22}(t)$
time
time


OTOC for α = 0.80, t_past = -20.0
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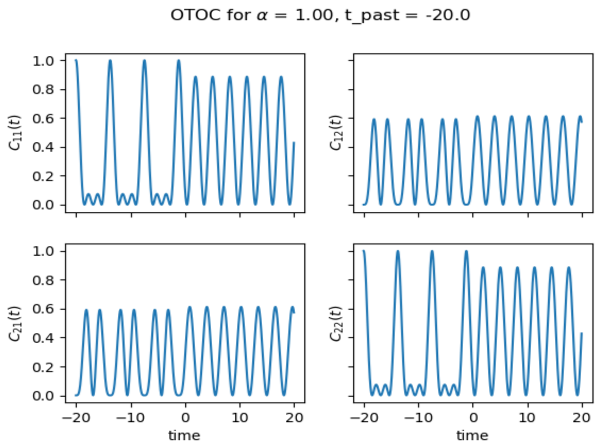
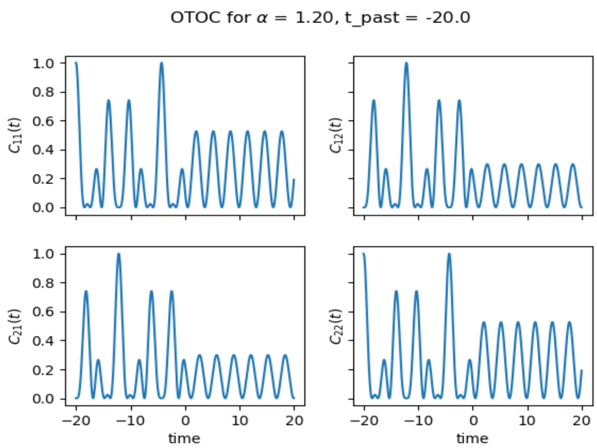
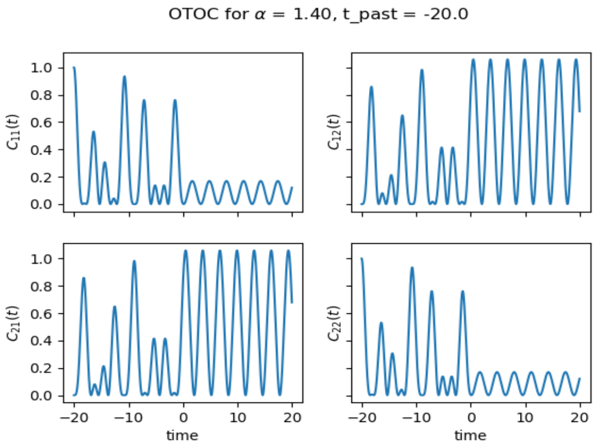
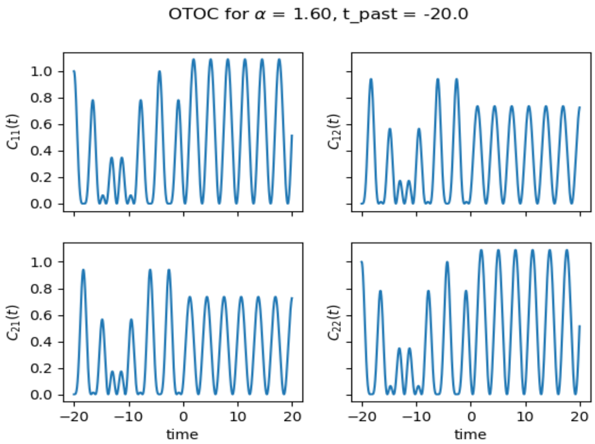
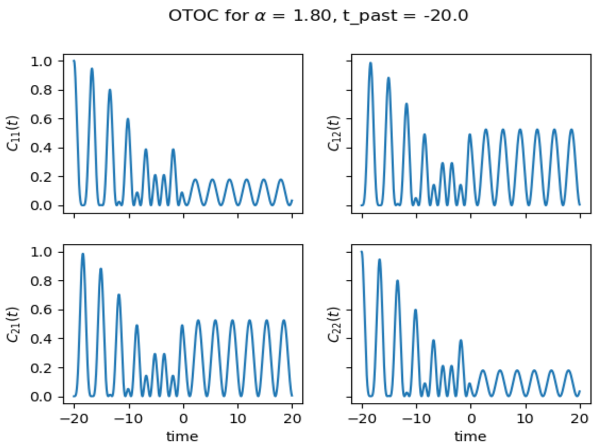
$C_{21}(t)$
$C_{22}(t)$
time

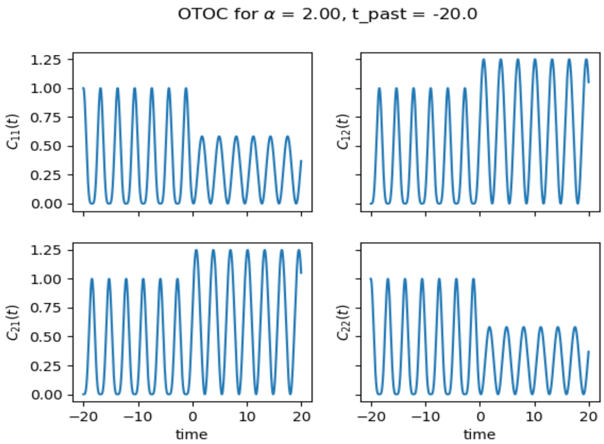

OTOC for α = 1.00, t_past = -20.0
$C_{11}(t)$
$C_{12}(t)$
$C_{21}(t)$
$C_{22}(t)$
time


OTOC for α = 1.20, t_past = -20.0
$C_{11}(t)$
$C_{12}(t)$
$C_{21}(t)$
$C_{22}(t)$
time


OTOC for α = 1.40, t_past = -20.0
$C_{11}(t)$
$C_{12}(t)$
$C_{21}(t)$
$C_{22}(t)$
time
time


OTOC for α = 1.60, t_past = -20.0
$C_{11}(t)$
$C_{12}(t)$
$C_{21}(t)$
$C_{22}(t)$
time
time


OTOC for α = 1.80, t_past = -20.0
$C_{11}(t)$
$C_{12}(t)$
$C_{21}(t)$
$C_{22}(t)$
time
time



10.4 Two Harmonic Chains

Consider a system with two separate Harmonic oscillator chains, one with the size N and the other M. The total hamiltonian of the system is given by,

$$H(t) = \sum_{k=1}^{N+M} \left(\frac{p_k^2}{2} + \frac{1}{2}\omega_0^2 x_k^2 \right) - \sum_{\substack{k=1 \\ k \neq N}}^{N+M} \omega_0^2 x_k x_{k+1} - \alpha(t) x_N x_{N+1} \tag{10.42}$$

Where,

$$\alpha(t) = \begin{cases} \omega_0^2, & \text{if } -\infty < t \leq 0 \\ 0, & \text{if } 0 < t < \infty \end{cases} \tag{10.43}$$

Denote the normal coordinates by X_k and the corresponding conjugate momenta by P_k. At $t \to -\infty$ all the oscillators are connected. They have the normal mode frequencies given by,

$$\bar{\omega}_k = 2\omega_0 \left| \sin \frac{\pi k}{2(N+M+1)} \right| \tag{10.44}$$

The Hamiltonian will become,

$$H(-\infty) = \sum_{k=1}^{N+M} \left(\frac{P_k^2}{2} + \frac{1}{2}\bar{\omega}_k^2 X_k^2 \right) \tag{10.45}$$

Transform the problem to a dimensionless one by dividing by ω_0. Then,

$$\tilde{H}(-\infty) = \sum_{k=1}^{N+M} \left(\frac{\tilde{P}_k^2}{2} + \frac{1}{2}\tilde{\omega}_k^2 \tilde{X}_k^2 \right) \tag{10.46}$$

Where $\tilde{H} = \frac{H}{\omega_0}$, $\tilde{\omega}_k = \frac{\bar{\omega}_k}{\omega_0}$, $\tilde{P}_k = \frac{P_k}{\sqrt{\omega_0}}$ and $\tilde{X}_k = \sqrt{\omega_0} X_k$.

Hence the ground state wave function is given by,

$$\psi_0^{(-\infty)}(\tilde{X}_i) = \prod_{k=1}^{N+M} \left(\frac{\tilde{\omega}_k}{\pi} \right)^{\frac{1}{4}} \exp\left\{ -\frac{\tilde{\omega}_k}{2} \tilde{X}_k^2 \right\} \tag{10.47}$$

At time $t = 0$ the two chains splits and will have different set of normal mode frequencies given by,

$$\left.\begin{aligned} \bar{\omega}_k^N &= 2\omega_0 \left| \sin \frac{\pi k}{2(N+1)} \right| \\ \bar{\omega}_k^M &= 2\omega_0 \left| \sin \frac{\pi k}{2(M+1)} \right| \end{aligned}\right\} \tag{10.48}$$

Then the Total Hamiltonian in the normal dimensionless coordinates will become,

$$\tilde{H}(0) = \sum_{k=1}^{N} \left(\frac{\tilde{P}_k^2}{2} + \frac{1}{2}(\tilde{\omega}_k^N)^2 \tilde{X}_k^2 \right) + \sum_{j=N+1}^{M} \left(\frac{\tilde{P}_j^2}{2} + \frac{1}{2}(\tilde{\omega}_j^M)^2 \tilde{X}_j^2 \right) \tag{10.49}$$

Then any general normalized eigenstate of the Hamiltonian is,

$$\begin{aligned} \psi_{[i]}^{(0)}(\tilde{X}_{[i]}) = \prod_{k=1}^{N} \prod_{j=N+1}^{N+M} \frac{1}{\sqrt{2^{m_j+n_k} m_j! n_k!}} \left(\frac{\tilde{\omega}_k^N \tilde{\omega}_j^M}{\pi^2} \right)^{\frac{1}{4}} \\ \times H_{n_k}(\sqrt{\tilde{\omega}_k^N} \tilde{X}_k) H_{m_j}(\sqrt{\tilde{\omega}_j^M} \tilde{X}_j) \exp\left\{ -\frac{1}{2}(\tilde{\omega}_k^N \tilde{X}_k^2 + \tilde{\omega}_j^M \tilde{X}_j^2) \right\} \end{aligned} \tag{10.50}$$

Where $[i]$ collectively denotes $N + M$ different indices the wave function has. As in the previous case, we need to write the ground state of $H(-\infty)$ as a linear combination of the eigenstates of $H(0)$. i.e,

$$\psi_0^{(-\infty)}(\tilde{X}_{[j]}) = \sum_{[i]} C_{[i]} \psi_{[i]}^{(0)}(\tilde{X}_{[j]}) \tag{10.51}$$

And,

$$C_{[i]} = \int \cdots \int d\tilde{X}_1 \cdots d\tilde{X}_N d\tilde{X}_{N+1} \cdots d\tilde{X}_{N+M} \; \psi_0^{(-\infty)} (\psi_{[i]}^{(0)})^* \tag{10.52}$$

As each of the integral is independent, we can write,

$$\begin{aligned} C_{[i]} = &\left[\prod_{k=1}^{N} \int_{-\infty}^{\infty} d\tilde{X}_k \frac{1}{\sqrt{2^{n_k} n_k!}} \left(\frac{\tilde{\omega}_k \tilde{\omega}_k^N}{\pi^2} \right)^{\frac{1}{4}} \exp\left\{ -\frac{1}{2}(\tilde{\omega}_k + \tilde{\omega}_k^N) \tilde{X}_k^2 \right\} H_{n_k}(\sqrt{\tilde{\omega}_k^N} \tilde{X}_k) \right] \\ \times &\left[\prod_{j=N+1}^{N+M} \int_{-\infty}^{\infty} d\tilde{X}_j \frac{1}{\sqrt{2^{m_j} m_j!}} \left(\frac{\tilde{\omega}_j \tilde{\omega}_j^M}{\pi^2} \right)^{\frac{1}{4}} \exp\left\{ -\frac{1}{2}(\tilde{\omega}_j + \tilde{\omega}_j^M) \tilde{x}_j^2 \right\} H_{m_j}(\sqrt{\tilde{\omega}_j^M} \tilde{X}_j) \right] \end{aligned} \tag{10.53}$$

Using the integral from 10.25, if all the m_j and n_k are even, then,

$$\begin{aligned} C_{[i]} = \prod_{k=1}^{N} \prod_{j=N+1}^{N+M} \frac{1}{\sqrt{2^{m_j+n_k} m_j! n_k!}} \left(\frac{\tilde{\omega}_k^N \tilde{\omega}_j^M \tilde{\omega}_k \tilde{\omega}_j}{\pi^4} \right)^{\frac{1}{4}} \frac{(-1)^{\frac{m_j+n_k}{2}} m_j! n_k!}{2^{\frac{m_j+n_k}{2}} (m_j/2)!(n_k/2)!} \\ \left(\frac{4\pi^2}{(\tilde{\omega}_k + \tilde{\omega}_k^N)(\tilde{\omega}_j + \tilde{\omega}_j^M)} \right) \times \left(1 - \frac{2\tilde{\omega}_k}{\tilde{\omega}_k + \tilde{\omega}_k^N} \right)^{\frac{n_k}{2}} \left(1 - \frac{2\tilde{\omega}_j}{\tilde{\omega}_j + \tilde{\omega}_j^M} \right)^{\frac{m_j}{2}} \end{aligned} \tag{10.54}$$

If any one of the $N + M$ indices is odd, then $C_{[i]} = 0$

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